

University of Groningen
Bachelor Thesis

Humpty Dumpty: Coherence Measure and QGEM Protocol

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Abstract

The 1988 papers from J. Schwinger under the same title, "Is spin coherence like Humpty-Dumpty?", gave an introduction and a general theory layout for the coherence measure of a $\frac{1}{2}$ -spin particle in a Stern–Gerlach Interferometer (SGI) and its main afferent complications. In this thesis, the particular case of a conserved x component of the spin is considered and all the steps which lead to his final result are retraced and detailed upon, from both a mathematical and physical perspective.

The context of this work is that of the Quantum Gravity Induced Entanglement of Masses (QGEM) protocol, which is an experimental proposal whose goal is to test the quantum nature of gravity by letting the spatially split states of two SGIs entangle through their gravitational interaction.

Given this background, a more general approach was taken, meant to illustrate what improvements can be brought to the magnetic field profiles used and the initial conditions, in order to maximize the system's coherence.

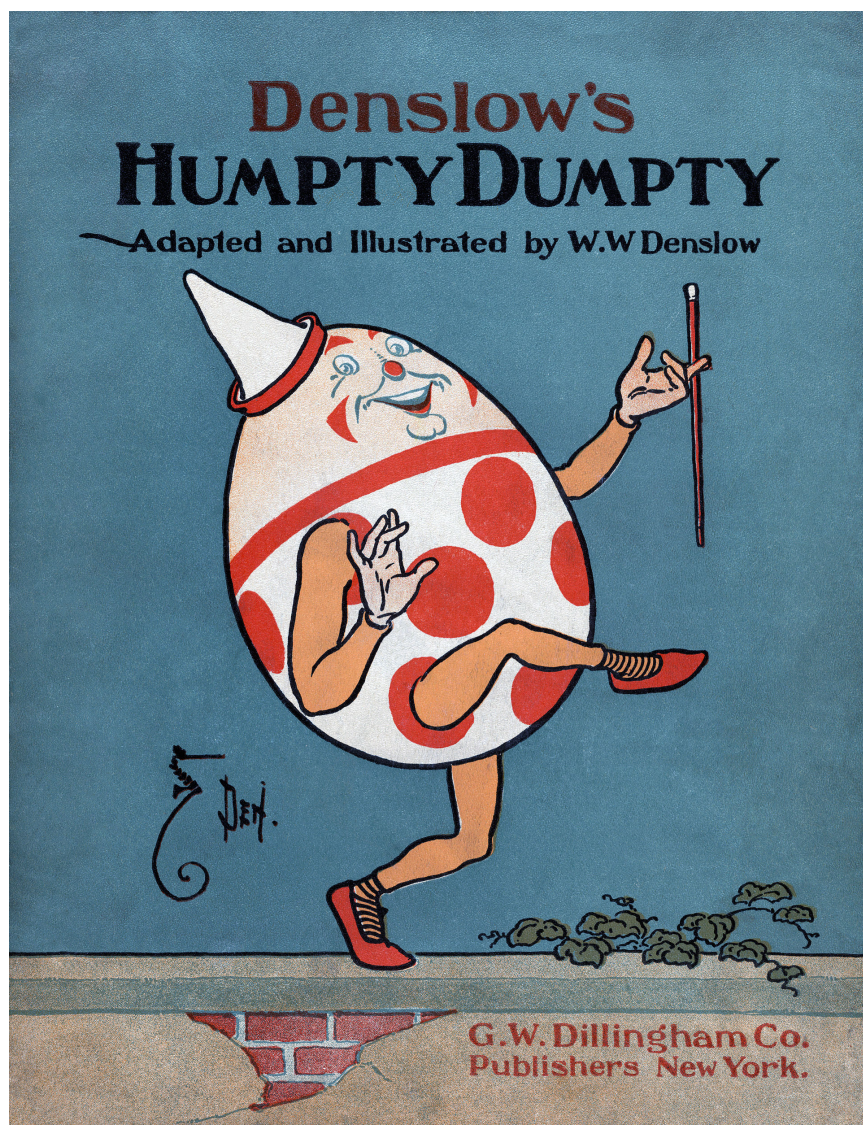


Figure 1: The 1904 cover of the adapted and illustration version of "Humpty Dumpty" by W. W. Denslow.

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Chapter 1 - Introduction and Historical Background

*Humpty Dumpty sat on a wall,
Humpty Dumpty had a great fall.
All the king's horses and all the king's men
Couldn't put Humpty together again.*

These are the verses of the famous English nursery rhyme "Humpty Dumpty" (HD) which J. Schwinger used to create one of the most excellent scientific metaphors with his work from 1988 [1], [2]. The answer to this riddle is a simple egg, showing how little control we humans have over even the simplest of phenomena.

The idea of putting HD back together seems plausible at first—if all it took was to place the atoms in a similar configuration—but J. Schwinger's HD demands a much different type of precision, dictated by the laws of Quantum Theory (QT). His papers [1], [2] describe a Stern–Gerlach Interferometer (SGI) spatially splitting along one axis a $\frac{1}{2}$ -spin particle, represented by Schrödinger's Gaussian, and subsequently recombining it. The success of this operation goes under the name of the coherence measure (C), and finding its expression is the primary goal of this thesis.

Of course, the working principle of the SGI is represented by the interaction of the spin's angular momentum with the magnetic field's gradient. On the two states, it exerts forces of equal magnitude in opposite directions along the magnetic field lines—along the x direction—where the spin's eigenvalues dictate the orientation. In **Chapter 4**, an extensive analysis of the magnetic field profiles is carried, resulting in reducing all of its components, apart from: B_x and $\frac{\partial}{\partial x}B_x$.

The challenge this quantum HD imposes comes from the uncertainties in momentum and position of the wave-packets used, as different parts on them probe the potential energy at different points in space [1], leading to a different phase gain inside of the packets. It accumulates to a total which represents the phase uncertainty, as discussed in **Chapter 3.2**.

Of course, one must account for all the changes in the potential when reconstructing the HD, such as fluctuations in the magnetic field and any external factors. One such fluctuation whose scientific importance cannot be understated is the gravitational one, and its detection is what has driven scientists for decades.

In the currently most widely accepted scientific framework, there are four fundamental forces of nature: the weak, strong, electromagnetic and gravitational. While the first three have been proven experimentally to have their own quantum mediators:

- strong field $\rightarrow$ gluons
- weak field $\rightarrow$ the $W^\pm$ and S^0 bosons
- electro-magnetic field $\rightarrow$ photons,

there is no such hard proof nor a theoretical need for the existence of the gravitational field's counterpart—the graviton—even while it perfectly fitting in the QT. Nevertheless, the importance of experimental proof for the existence of the graviton is enormous, not only for the completeness of the theory but also for the further indications it would bring to the limits of QT and their implications on cosmological scales [3]. Such effects include violations of Lorentz invariance [4] and imprints of quantum gravitational effects in the polarization of the Cosmic Microwave Background (CMB) [5].

Most of the proposed experimental protocols rely on direct, extremely fine measurements of the field's classical effect or even a pure detection of a graviton [6]. These approaches require tremendous amounts of precision, even by today's standards, and seem to lead to a dead-end for the foreseeable future.

The most promising alternative is that of the Quantum Gravity induced Entanglement of Masses (QGEM) protocol. It tests whether gravity is classical or quantum in a table-top experiment, as described by S. Bose and A. Mazumdar in [7]. It is based on setting two SGIs next to each other and observing if entanglement is produced between the (two) closest spatially split states (one from each SGI). A more detailed discussion is provided in **Chapter 2**, where factors such as the entanglement of states and the core principles of the protocol are laid down, together with other considerations such as the obtaining of Schrödinger's Gaussian in the real-life experiment.

In **Chapter 3**, the problem of better understanding the C and its interactions with the spin components is better defined through a simplified treatment of a $\frac{1}{2}$ -spin particle in a homogenous magnetic field.

The main part of this thesis resides in **Chapters 5** and **6**. The former begins by providing a definition of the system's Hamiltonian and making the transition from Schrödinger's picture to the Hermitian one. This permits the determination of the system's equations of motion and a general form of the evolution operators. It also provides a clearer redefinition of the magnetic field's profile which accounts for its interaction with the two states. In the later chapter, the exact form of the evolution operators is given, allowing for a direct and thoroughly explained achieving of the final form of C utilizing the expectation value of the ladder operator S_+ . Other effects such as the Longitudinal Stern–Gerlach (LSG) effect are considered.

In **Chapter 8**, a more complex magnetic field profile is considered for the sake of obtaining a different perspective of the previous findings from both physical and mathematical standpoints.

A discussion on the protocol, its limitations, and possible improvements are provided together with the relevant literature in the **Discussion**, together with clarifications of numerous principles and terms used.

Chapter 2 - The QGEM Protocol

2.1 Complete Experimental Setup

In the Stern–Gerlach effect, a magnetic field gradient splits particles into spatially separated paths according to their spin projection. The idea of exploiting this effect for creating coherent momentum superpositions for matter-wave interferometry appeared shortly after its discovery, almost a century ago, but was judged to be far beyond practical reach [?].

The QGEM protocol from the original paper [7] is based on two SGIs, each spatially splitting the two states of one $\frac{1}{2}$ -spin state particle (whose mass is denoted by m_1 and m_2), as depicted in **Fig.2.1**.

The test masses are held steadily in this superposition of their two respective spatially separated spin states—*down* $|\downarrow\rangle_{1,2}$ and *up* $|\uparrow\rangle_{1,2}$ —for a total time T .

In an ideal SG, the two separation of the beams ($\Delta x_{1,2}$) is large enough such that the two states do not overlap ("good" spatial splitting):

$$\langle \downarrow | \uparrow \rangle_{1,2} = 0, \quad (2.1.1)$$

where the index is common for the two states.

In reality, of course, the overlap cannot be zero, but for all tens and purposes of this thesis and this protocol (using parameters such as those from [8] and [9], this approximation is more than appropriate. In order for the achieve this spread, the macroscopic (Δ) and microscopic (δ) ones in momentum and position must respect the inequalities (during the splitting and recombination stages):

$$\Delta p_x(t) \gg \delta p_x \quad | \quad \Delta x(t) \gg \delta x \quad | \quad \Delta p_x(t) \Delta x(t) \gg 1, \quad (2.1.2)$$

that is, the macroscopic splittings of position and momentum are much larger than the width of the wave-packets and their Fourier transform, respectively. The reason behind the omission of the time dependence of the microscopic spreads,

$$\delta p_x(t) = \delta p_x \quad | \quad \delta x = \delta x(t'), \quad (2.1.3)$$

for any time t , is that the wave-packets are assumed to be held in a harmonic potential with adiabatic changes [10]. Thus, they maintain their shape, i.e., there is no spread. The values of its position and momentum uncertainties oscillate through the potential, maintaining a saturated minimum uncertainty bound.

However, for a complete treatment, one must consider the times when such a potential cannot be exerted—only the momentum spread can remain constant. Even when no initial correlation between δx and δp_x is present, the former has a time dependence given by (see the **Appendix A** for the complete proof):

$$(\delta x(t))^2 = (\delta x)^2 + \left(\frac{t}{m}\right)^2 (\delta p_x)^2, \quad (2.1.4)$$

implying a moderate spatial spread of the beams, achieved for a total duration of the order:

$$T \sim \frac{m \delta x}{\delta p_x}. \quad (2.1.5)$$

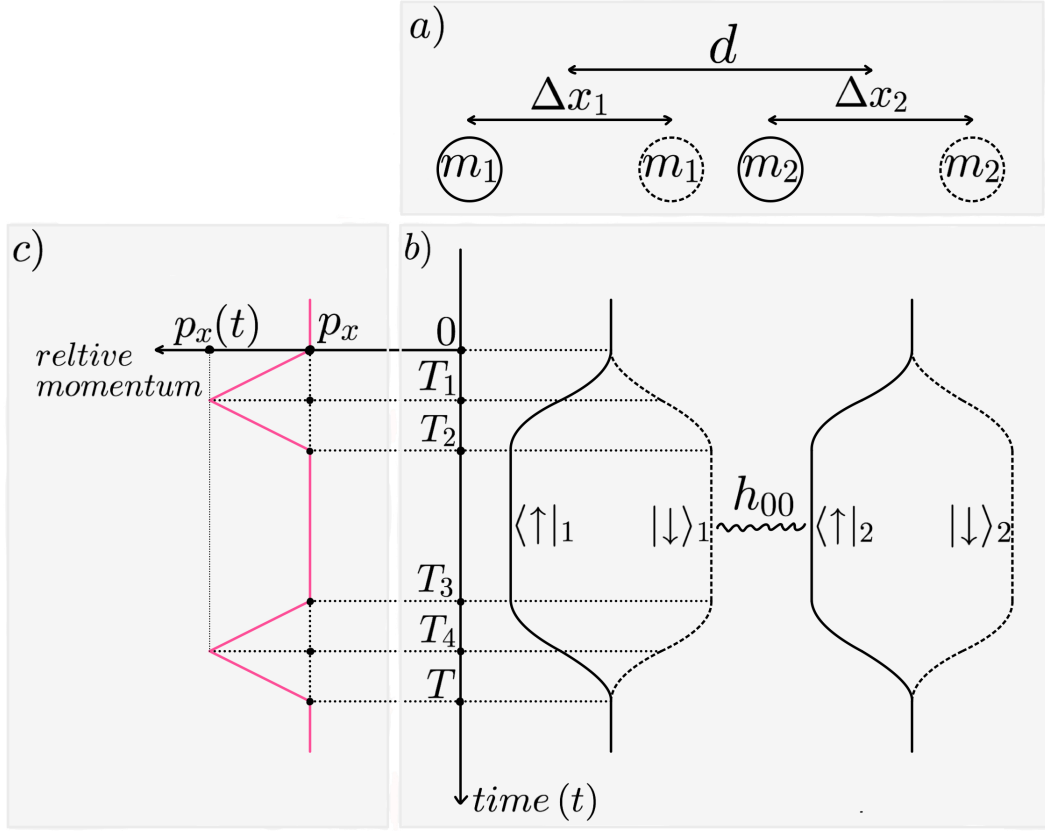


Figure 2.1: Setup of the current proposition of the QGEM-protocol, as depicted in [7]. It is comprised of three parts denoted by the grey boxes:

- a)** The separations between the average trajectories of the two particles (d) and that of their spatially split states due to the SGIs (Δx_1 and Δx_2).
- b)** Scheme of the complete apparatus, comprising of two adjacent SGIs and the trajectories of the states, as represented by the two arms, respectively. The time scale depicts the main stages of the apparatus. The evolution under mutual gravitational interaction (h_{00}) during the time interval $[T_2, T_3]$, entangling the two test masses by imparting appropriate phase to the components of the superposition.
- c)** The macroscopic momentum for the SGI's states, relative to its value at time $t = 0$, according to constant gradients in each phase.

By setting the two closest arms of the SGIs very close to each-other at a separation distance $d - \Delta x$, one maximizes the entanglement between the four states, which, given an original ($t = 0$) combined state:

$$|\Psi(t=0)\rangle_{12} = \frac{1}{\sqrt{2}}(\langle | \uparrow \rangle_1 + | \downarrow \rangle_1) \cdot \frac{1}{\sqrt{2}}(\langle | \uparrow \rangle_2 + | \downarrow \rangle_2), \quad (2.1.6)$$

result in the final ($t = T$) state:

$$|\Psi(t=T)\rangle_{12} = \frac{e^{i\phi}}{\sqrt{2}} \left[\langle | \uparrow \rangle_1 \frac{1}{\sqrt{2}} (\langle | \uparrow \rangle_2 + e^{i\Delta\phi_{du}} | \downarrow \rangle_2) + | \downarrow \rangle_1 \frac{1}{\sqrt{2}} (e^{i\Delta\phi_{ud}} \langle | \uparrow \rangle_2 + | \downarrow \rangle_2) \right]. \quad (2.1.7)$$

In the above equation, the exponential phase factors represent the different interactions of the four superposition components. They account for their respective gravitational interaction energies dictated by their (distinct) separations and are given by:

$$\Delta\phi_{ud} = \phi_{du} - \phi \quad | \quad \Delta\phi_{ud} = \phi_{ud} - \phi \quad (2.1.8)$$

$$\phi_{ud} \sim \frac{Gm_1m_2T}{d - \Delta x} \quad | \quad \phi_{ud} \sim \frac{Gm_1m_2T}{d + \Delta x} \quad | \quad \phi \sim \frac{Gm_1m_2T}{d}, \quad (2.1.9)$$

where G is the gravitational constant.

Of course, one must first assure that the produced entanglement originates only from the gravitational interaction and not the weak, strong, or electromagnetic. The first two can be dealt with quickly by setting a minimum for the separation distance [7]. The latter, however, is more tricky, as, just like the gravitational interaction, it has infinite range. The solution is to place a very thin, very conducting metallic layer between the two SGIs, which acts as a Faraday cage screening the electromagnetic interaction between the two superposed masses.

This, however, gives rise to another problem: there is now a Casimir-Polder (CP) potential between the conducting plate and the individual superposed masses [7], leading to an acceleration of the inner states $|\downarrow\rangle_1$ and $|\uparrow\rangle_2$ towards the plate during the free-fall stage $[T_2, T_3]$.

While an average of this effect can be measured using atomic interferometry [11], it must overall be reduced to a tenth of the strength of the gravitational potential between the two states in order to achieve a detectable entanglement phase [7].

2.2 Entanglement: LOCC and LOQC

The QGEM protocol is based around a single assumption:

The gravitational interaction between two bodies is mediated by a gravitational field—that is, it is not a direct interaction-at-a-distance.

While this concept is reminiscent of Newton's "cosmic catastrophe" of a disappearing Sun—for which the bodies orbiting it would *instantaneously* change their trajectories—it takes a much stronger meaning than this simple relativistic action-at-a-distance problem by making use of the local operators and classical communication (LOCC) method and its counterpart, local operators and quantum communication (LOQC). This is the core of the protocol.

This non-interacting-at-a-distance nature of the gravitational fields represents the local operator (LO) terms in the two principles and is, in fact, characteristic of Quantum Field Theory (QFT) (local in comparison to the separation of the masses [12]) and thus cannot be circumvented.

The classical communication "CC" and quantum communication "QC" parts imply that quantum information—such as a qubit—can *only* be transmitted via a quantum bus [12]. That is to say:

If the gravitational field is classic in nature, the spins of the two particles cannot become entangled through it.

Thus, any possible entanglement is done through LOQC, with the quantum channel being an off-shell/virtual graviton [12]. The resulting growth of entanglement in a pure state of the two masses follows a $\frac{1}{r}$ law. One physical way to visualizing this process is by comparison to trapped ion quantum computation, where a quantum bus—namely, a quantized vibrational mode—is necessary for the entanglement of two ions [8].

Put into the context of **Eq.2.1.7**, for no entanglement, the two terms in the square brackets are now equal and thus, the whole expression can be factored out to the form in **Eq.2.1.6**. One should be very careful when constructing the setup, as the same result can be obtained in the special case:

$$\Delta\phi_{du} + \Delta\phi_{du} = 2n\pi, \quad (2.2.1)$$

for any integer n . In fact, the entanglement between the masses increases monotonically over $\Delta\phi_{du} + \Delta\phi_{ud}$ as it evolves from 0 to π , with maxima obtained at π [8].

2.3 Gaussian

Given that the scope of this thesis is the analysis of the recombination of the two spatially split states in each SGI, only the mechanism of a single one is considered—as they are set to work in an identical manner—and thus drop the suffixes 1 and 2 seen in **Fig.2.1**. Accordingly, the interactions of the complete apparatus (h_{00} term in **Fig.2.1**) result in a change of the potential felt by the two states [13] and are beyond the scope of this thesis.

The two initial ($t = 0$) states consist of gaussian wave-packets, by curtesy of using a Bose Einstein Condensate (BEC) as probe [9], [7]. Such a system follows the minimum uncertainty wave function [14]:

$$\Psi_0^G(x) = (2\pi)^{-\frac{1}{4}}(\delta x)^{-\frac{1}{2}}e^{i\langle p_x \rangle(x - \langle x \rangle)}e^{-\frac{1}{4}\left(\frac{x - \langle x \rangle}{\delta x}\right)^2}, \quad (2.3.1)$$

also known as the Schrödinger's Gaussian. The time dependence of $x(t)$ is set to represent the reference frame of the states (as they move through the potential). This notation is explained and argued in favor of in **Chapter 4.2** and **4.3**, with a more robust reasoning from QT perspective in **Chapter 5.4.2**. Another restriction on the general discussion is the focus on the x component of the field and wave function, which is explained in the same sections. For now, one should not focus on them too much, as the mathematics and physical interpretations do not change from the general case, which is still represented first, where necessary.

A very special property of this Schrödinger's Gaussian is that it is defined to saturate the Minimum Uncertainty Principle (MUP) in equal proportions by the position and momentum components:

$$\delta p_x = \delta x = \frac{1}{2}, \quad (2.3.2)$$

despite the wave packet in **Eq.2.3.1** staying normalized for any other position and momentum uncertainties which still saturate this bound. Such cases are coined as squeezed states and are of great help in the QGEM protocol (more details in **Chapter 7**).

2.4 Choice of Axis

To simplify the problem, the visualization and argumentation of the magnets' setting in the apparatus and discussing the importance of all of the magnetic field components are imperative.

As seen in **Fig.2.2**, the axis system is right-handed and has undergone a $\frac{\pi}{2}$ clockwise rotation around the z axis, convention used to preserve the familiar form of the Pauli spin matrices (**Eq.3.1.1**). The z axis was chosen as the vertical one (opposite to the direction of the gravitational attraction exerted by the Earth), reminding that the experiment must be carried in a free-fall environment such as a free-fall tower or on the International Space Station (ISS) [7], [13]. Obviously, this is done to discard the z dependency of the magnetic field caused by Earth's gravitational attraction.

The precision and large gradients of the magnetic fields required by this protocol [7] can only be approached simultaneously by atom chips [15]. Given the convention above, their positioning must be parallel to the y - z plane, as depicted in **Figure 2.2** by the yellow quadrupole field formation—which they must form.

This field profile is the result of the three-wire configuration required for the reduction of the hindering effects of the field, which also has the benefit of exposing the wave-packets to a weaker magnetic field while maintaining strong gradients [9], [15]. The main advantages of this method are the low-phase noise (standard deviations (SD) as low as 0.1 rad of the spatial fringe SGI), demonstrating the stability of the apparatus.

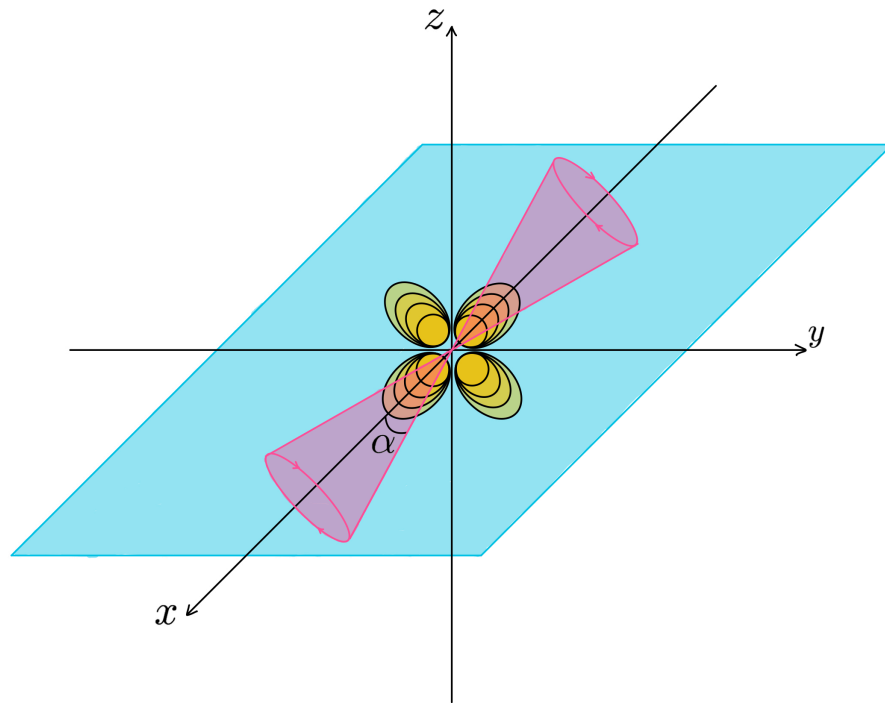


Figure 2.2: Scheme of the system of axis used and the orientation of the spin (depicted as two pink cones along the x axis, each representing one of its states) and atom chip (represented in yellow by its quadrupole field). The x - y plane (represented in blue) is chosen for containing the only gradient components of the field—therefore being the only plane in which the states gain momentum. The opening angle of the cones is denoted by α .

Chapter 3 - Defining the Problem: Coherence

3.1 Spin

In order to advance further, the basis of the spin problem must now be laid down.

From the particular orientation of the spin and of the axis (pointing in the x direction as in **Fig.2.2**), the Pauli spin matrices are now defined as [16]:

$$\sigma_x = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} \quad \Bigg| \quad \sigma_y = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \quad \Bigg| \quad \sigma_z = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}. \quad (3.1.1)$$

The imbedded spin angular momentum and spin magnetic moment of the particle are denoted by $\vec{S}$ and μ , respectively, and are related by (maintaining the notation from [2]):

$$\vec{\mu} = 2\gamma\vec{S} \quad \Bigg| \quad \vec{S} = \frac{1}{2}\vec{\sigma} \quad \Bigg| \quad \vec{\mu} = \gamma\vec{\sigma}, \quad (3.1.2)$$

where γ is the gyromagnetic ratio.

When presented to a homogenous magnetic field pointing in along in the same x direction, the spin's x component remains constant, while the y and z ones precess around the x axis. In the quantum language, the x component has two good discrete eigenvalues that form a superposition ($\mp\frac{1}{2}$, representing the *down* and *up* states), while the other two spin components are in a superposition with each-other, precessing with the Larmor frequency:

$$\omega(t) = -\gamma\vec{B}(t) \quad \Bigg| \quad \omega_x(t) = -\gamma B_x(t), \quad (3.1.3)$$

where $\vec{B}(t)$ is the magnitude of the homogenous magnetic field felt by the particle at time t and $B_x(T)$ is its component in the x direction (this notation makes use of the same argument as the Gaussian wave function from the previous chapter). If the charge of the particle is negative, the precession is counter-clockwise, as depicted in **Fig.2.2**.

One should be warned that the time dependence of the magnetic field (and, by extension, of the force) represents the movement of the states through the field from their frame of reference and not an actual physical time variation of the fields. This notation becomes clear in due course, but it is imperative to firstly indicate clearly the main direction of this thesis.

As the spin precesses around the x axis, one finds useful the following non-Hermitian combination:

$$S_+ = S_y + iS_z = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}, \quad (3.1.4)$$

representing the (raising) spin ladder operator. As an argument for its non-Hermiticity, S_y and S_z are both Hermitian and so is the sum of any two Hermitian operators. However, the product of a Hermitian operator with a complex constant is Hermitian only if the constant is real, which i is not. Thus, S_+ is not Hermitian.

By looking again at **Fig.2.2**, one suspects the expectation value of S_x to remain constant throughout the splitting (as it has two "good" eigenvalues), while those of S_y and S_z to average to zero if integrated over long time periods, due to their precession around the x axis—that is, their expectation values do not result in null values.

It is still unclear how the Ladder operator is used in the context of the HD problem. To reach an answer to this, the problems of what the coherence measure is and how it can be obtained must first be answered. Thus, its use and necessity are described and argued for in **Chapter 3.3**.

3.2 Wave Function Splitting in SG: Humpty Dumpty

If the SG's magnetic field is turned on, its two states (*up* and *down*) are subjected to a force:

$$\vec{\mathcal{F}}(t) = \vec{\nabla}(\vec{\mu} \circ \vec{B}(\vec{r}(t))) = \gamma \sigma_x \frac{\partial}{\partial x} B_x(t) \hat{x} \quad \Bigg| \quad \mathcal{F}_x(t) = \gamma \frac{\partial}{\partial x} B_x(t), \quad (3.2.1)$$

where in the first equation, the gradient and the Hadamard product were used ($\vec{\nabla}$ is the vector differential operator). The last equality has been obtained by restricting the magnetic field and its gradient to their x components. The matrix term σ_x has also been reduced as it only represents the two states, which experience forces of equal magnitude but opposite sign (thus the splitting parallel to the field).

In order to satisfy the spatial separation of the states—no overlap of the spatially split states' wave functions **Eq.2.1.1**—required by good SG and SGI apparatus, the macroscopic and microscopic spreads in momentum and position must respect **Eq.2.1.2**. Such a macroscopic effect can be achieved if the following is satisfied:

$$\int_0^{T_2} \mathcal{F}_x(t) dt = \Delta p_x(T_2) \gg \delta p_x, \quad (3.2.2)$$

where the time T_2 was chosen in consideration of **Fig.2.1**. This requirement of the force only has to be satisfied by its x component, as the (main) splitting of the states is along the same axis.

The durations of the splitting and recombination of the SGI ought not to be the same, due to effects such as the Longitudinal SG Effect (see **Chapter 6.7**). Thus, the splitting and recombinations stages only have approximately equal durations:

$$T_2 \approx T_5 - T_3. \quad (3.2.3)$$

As mentioned in the **Introduction**, the force $\vec{\mathcal{F}}(t)$ acts on each point of the wave-packets individually. Thus, at time T_2 , a total phase inside of it is introduced:

$$\frac{1}{2}\Phi(T_2) = \frac{1}{2} \left[- \int_0^{T_2} 2\gamma B(t) dt \right] = - \int_0^{T_2} \gamma B(t) dt = - \int_0^{T_2} \omega(t) dt, \quad (3.2.4)$$

where $\Phi(t)$ is the Larmor angle. Its variation is given by:

$$\begin{aligned} \delta\Phi(T_2) &= -\gamma \int_0^{T_2} \delta\vec{B}(t) dt = -\gamma(\delta\vec{r}) \circ \int_0^{T_2} \vec{\nabla}\vec{B}(t) dt = -\gamma(\delta\vec{r}) \circ \int_0^{T_2} \vec{\mathcal{F}}(t) dt \quad \Bigg| \quad \delta\Phi_x(T_2) = -\gamma\delta x \int_0^{T_2} \frac{\partial}{\partial x} B_x(t) dt \\ &= -\delta x \cdot \Delta p_x(T_2) \\ &\gg -\delta x \cdot \delta p_x = -\frac{1}{2}, \end{aligned} \quad (3.2.5)$$

where for the first inequality, **Eq.3.2.2** was used.

One must emphasize the following aspect of this equation: it does not represent a relative phase between the two beams but rather a dispersion of the phases within each individual beam. It is this phase dispersion that must be reversed during the refocusing of the beam in the second stage of the SGI and requires an extreme amount of precision and fine-tuning of the apparatus (beyond what is necessary for the spatial refocusing).

Obviously, the left-hand side of the above equation can be set to be much larger ($\gg$) than a vector with only $\frac{1}{2}$ entries, but, based on the argument which followed **Eq.3.2.2**, it does not have to. However, given the use of the Gaussian wave-packet, there are still natural packet spreads on which the $\vec{\mathcal{F}}(t)$ acts upon in the y and z directions, so the vector thus formed does not have any null entry. This directly results in a spread of the packet, solely conditioned by the existence of a magnetic field gradient in the respective direction.

This spread is encountered again in **Chapter 6.3**, in the full mathematical description of C .

3.3 Humpty Dumpty and Coherence Measure

In general, two wave sources are defined as coherent if their frequency and waveforms are identical. Put into the context of the SGI and HD, the waveforms of the two spatially split states undergo a change due to their finite size and the uncertainties they have. Such effects are those of the phase shift inside the position and momentum wave-packets. Also, there are macroscopical effects that must be accounted for and reversed, such as the LSG effect from **Chapter 6.7**, the attractive CP potential, and even external factors during the experiment such as residual particles in the vacuum.

All of these form a combined effect which leads to a very hard experimental recombination of the beams, with both macroscopical and microscopical corrections to be taken into account. This is the HD problem in its full glory.

But how can C be measured and—thinking back to **Chapter 3.1**—how does the ladder operator S_+ provide any help in the matter?

Unfortunately, a complete mathematical reasoning for the need of the non-Hermitian combination requires equations not derived yet, which provide a much deeper understanding of the mathematical constructs at hand. However, the physical interpretation is a completely different matter and can be discussed right away.

Consider a $\frac{1}{2}$ -spin particle which, at time $t = 0$ is in a selected eigenstate of σ_y with eigenvalue $\sigma'_y = 1$ such that the spin's components have the following expectation values:

$$\langle 2S_y \rangle = 1 \quad | \quad \langle 2S_z \rangle = 0 \quad | \quad \langle 2S_x \rangle = 0. \quad (3.3.1)$$

The third equation directly implies that, upon subsequent measurement of σ_x , the beam is split in two parts, corresponding to the two states of the spin in the z direction. This translates to the (total) spin expectation value:

$$\langle 2\vec{S} \rangle = \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix} = 1, \quad (3.3.2)$$

where 1 denotes a unit vector, which in this case points in the y direction.

If, after time T , the spin's expectation value is still of unit length:

$$\langle 2\vec{S}(T) \rangle = 1, \quad (3.3.3)$$

then one of its components has the same +1 eigenvalue. This is the case of perfect coherence maintenance: Humpty Dumpty can smile again.

Unfortunately, one already knows how that song and dance goes, and such a realization in the case of the SGI is practically unattainable. Thus, the length of the vector is smaller than unity, showing a loss of coherence C .

As mentioned at the beginning of this chapter, one expects the spin's component in the x direction to maintain the same null expectation value at any time, due to the adiabatic changes the states undergo in the potential—thus maintaining

its eigenstates [10]. For the sake of demonstration and intuition, a simplified treatment where no SGI apparatus being utilized is in order.

Consider the $\frac{1}{2}$ -spin particle from above in a homogenous magnetic field pointing in the x direction ($B_x(t)$). Its Hamiltonian and spinor are given by:

$$H = -2\gamma B_x S_x = -\gamma B_x \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} \quad \left| \quad \chi(t) = \begin{pmatrix} \cos\left(\frac{\alpha}{2}\right) e^{i\Phi_x(t)} \\ \sin\left(\frac{\alpha}{2}\right) e^{-i\Phi_x(t)} \end{pmatrix}, \right. \quad (3.3.4)$$

where α is the angle at which $\vec{S}$ is tilted w.r.t. the x -axis, as shown in **Fig.2.2**. Thus, one can directly obtain the following expectation values of the spin's components [17]:

$$\langle \vec{S}(t) \rangle = \chi^\dagger(t) \vec{S} \chi(t). \quad (3.3.5)$$

The following forms of the spin's components' expectation values are directly obtained:

$$\begin{aligned} \langle 2S_y(t) \rangle &= \sin(\alpha) \cos(\Phi(t)) & \langle 2S_z(t) \rangle &= -\sin(\alpha) \sin(\Phi(t)) & \langle 2S_x(t) \rangle &= \cos(\alpha) \\ &= \cos(\Phi(t)) & &= -\sin(\Phi(t)) & &= 0, \\ &= \text{Re}\{e^{-i\Phi(t)}\} & &= \text{Im}\{e^{-i\Phi(t)}\} & & \end{aligned} \quad (3.3.6)$$

where a is the constant of proportionality of χ_+ in the composition of $\chi(t)$ and is related to the angle α by:

$$\alpha = 2 \cos^{-1}(a). \quad (3.3.7)$$

The time dependence of $\langle S_y(t) \rangle$ and $\langle S_z(t) \rangle$ becomes clear, representing their vectorial counterparts' precession around the x axis with the Larmor frequency given by **Eq.3.1.3**. However, there is, as expected, no such dependence for $\langle S_x(t) \rangle$.

In going from the first to the second line of the three equations composing **Eq.3.3.6**, the expectation values from **Eq.3.3.1** were used. Next, the third equation must return zero, thus the value of $\pi(n + \frac{1}{2})$ was found for α , where n is an integer. This lead to the reduction of the sin term from the first two equations.

For the next step in the remaining two equations of **Eq.3.3.6**, the complex form was adopted, form which is found again in the final form of C (apologies are in order to anyone who hates spoilers).

The three equations form a unit vector with its y and z components precessing around the x axis with the Larmor frequency. This is how the spin of the SGI should look like for a perfect rebuilt of the HD: an example of perfect spin coherence as the (total) spin expectation value $\langle S(t) \rangle$. This can also be seen by constructing the ladder operator $S_+(t)$:

$$\langle S_+(t) \rangle = \langle S_x(t) + iS_y(t) \rangle = \text{Re}\{e^{-i\Phi(t)}\} + i\text{Im}\{e^{-i\Phi(t)}\} = e^{-i\Phi(t)}, \quad (3.3.8)$$

and taking its modulus, which now represents the length of the vector $\vec{S}(t)$ —its expectation value:

$$\langle \vec{S}(t) \rangle = |\langle S_+(t) \rangle| = |e^{-i\Phi(t)}| = 1. \quad (3.3.9)$$

It is time to ask the big question directly: how does C act on the above expectation values?

It represents how "well" the recombined wave-packets overlap, accounting for their now different phases (shapes) and relative alignment, being defined to range from 0 to 1, thus reducing the length of the (originally) unitary vector of the total spin's expectation value. It is thus given by the following ratio:

$$\frac{\langle \vec{S}(T) \rangle}{\langle \vec{S} \rangle} = C, \quad (3.3.10)$$

but this directly implies that:

$$C = \frac{\langle \vec{S}(T) \rangle}{\langle \vec{S} \rangle} = \frac{|\langle S_+(T) \rangle|}{|\langle S_+ \rangle|} = \frac{|Ce^{-i\Phi(T)}|}{1}, \quad (3.3.11)$$

from which:

$$\begin{aligned} \langle S_+(T) \rangle &= Ce^{-i\Phi(T)} & \left| \quad \langle S_y(T) \rangle &= \text{Re}\{Ce^{-i\Phi(T)}\} \right. & \left. \langle S_z(T) \rangle &= \text{Im}\{Ce^{-i\Phi(T)}\} \right. \\ & & &= C\text{Re}\{e^{-i\Phi(T)}\} & &= C\text{Im}\{e^{-i\Phi(T)}\}. \end{aligned} \quad (3.3.12)$$

As a confirmation of the result, because C represents the residual phase differences in the wave-packets accumulated in the SGI and thus present in the wave functions used to obtain the expectation values above, it must be common throughout the expectation values.

One might now begin to question why the (composite) S_+ had to be chosen if C can be obtained only from the expectation value of S_y using the same division of its two expectation values method, but the answer requires a much stricter mathematical description of the phenomena. Thus, it is found in the **Discussion**.

This seems like the perfect point to start truly chasing after C , and to do so, the definition of the magnetic field profiles are the first step in line.

Chapter 4 - Magnetic Field Profile: Definition and Approximations

4.1 Stages

In order for spatial splitting and recombination of the states to be produced in the SGI, it must be inhomogeneous in that direction, i.e., its gradient in the x direction must not be zero.

As seen in **Fig.2.1**, one can divide the experimental procedure in five stages:

- A positive gradient — accelerating the two states in opposite directions — Splitting ($0 < t \leq T_1$);
- A negative gradient — decelerating the two states — Splitting ($T_1 < t \leq T_2$);
- No gradient — Free-fall ($T_2 < t \leq T_3$);
- A negative gradient — accelerating the two states in the same direction — recombination ($T_3 < t \leq T_4$);
- A positive gradient — decelerating the two states — recombination ($T_4 < t \leq T$).

While, obviously, each of these requires a different magnetic field profile, leading to an overall time-dependent one, the individual profiles can be time-independent. However, for a complete description, the approximation of an idealistic step function representation of those profiles must be accounted for. This can be done through a switching function, such as that in [13]:

$$S_W(t, t_{on}, t_{off}) = \frac{1}{2}[\tanh(\delta \cdot (t - t_{on})) + 1] \cdot \frac{1}{2}[\tanh(\delta \cdot (t_{off} - t)) + 1], \quad (4.1.1)$$

where δ is the switching frequency parameter and the two parameters t_{on} and t_{off} represent the ends of the time interval when the respective profile it acts on (being multiplied by it) is in use. δ must be sufficiently slow such that the magnetic field change is adiabatic at all times, while still maintaining the general (step) shape.

The parameters of such a function are dictated by the specific probe and the parameters which are desired to be maximized [13]. Regardless of that, the use of this function is beyond the scope of this thesis.

Given the above, in this thesis, a single time-independent magnetic field profile is used, with the generalization procedure and details being found in the **Discussion**. This includes any fluctuations the field might have, such as external factors, manufacturing imperfections, and the flow of the individual electrons through the atom chip.

4.2 Magnetic Field

As the two states have a non-zero macroscopic velocity with respect to the laboratory frame in the first and last two stages of the experiment and microscopic ones throughout it, in their reference frame, the magnetic field profiles are not constant. Thus, the time dependence of the position argument of the field is introduced (this notation is cumbersome due to its length, but, for the sake of completeness and consistency, it is kept). One should note the difference between the two:

$$\vec{B}(\vec{r}, t) \neq \vec{B}(\vec{r}(t)). \quad (4.2.1)$$

The magnetic field has three components (one for each direction) with its gradient having nine (three for each direction):

$$\vec{B}(\vec{r}(t)) = \begin{pmatrix} B_x(\vec{r}(t)) \\ B_y(\vec{r}(t)) \\ B_z(\vec{r}(t)) \end{pmatrix} \quad \left| \quad \vec{\nabla} \vec{B}(\vec{r}(t)) = \begin{pmatrix} \frac{\partial}{\partial x} B_x(\vec{r}(t)) & \frac{\partial}{\partial y} B_x(\vec{r}(t)) & \frac{\partial}{\partial z} B_x(\vec{r}(t)) \\ \frac{\partial}{\partial x} B_y(\vec{r}(t)) & \frac{\partial}{\partial y} B_y(\vec{r}(t)) & \frac{\partial}{\partial z} B_y(\vec{r}(t)) \\ \frac{\partial}{\partial x} B_z(\vec{r}(t)) & \frac{\partial}{\partial y} B_z(\vec{r}(t)) & \frac{\partial}{\partial z} B_z(\vec{r}(t)) \end{pmatrix}, \quad (4.2.2)$$

where the second equation represents the Jacobian of the field. The second and fourth Maxwell's equations are given by:

$$\vec{\nabla} \cdot \vec{B}(\vec{r}(t)) = \frac{\partial}{\partial x} B_x(\vec{r}(t)) + \frac{\partial}{\partial y} B_y(\vec{r}(t)) + \frac{\partial}{\partial z} B_z(\vec{r}(t)) = 0 \quad (4.2.3)$$

$$\vec{\nabla} \times \vec{B}(\vec{r}(t)) = \left[\frac{\partial}{\partial x} B_y(\vec{r}(t)) - \frac{\partial}{\partial y} B_x(\vec{r}(t)) \right] + \left[\frac{\partial}{\partial y} B_z(\vec{r}(t)) - \frac{\partial}{\partial z} B_y(\vec{r}(t)) \right] + \left[\frac{\partial}{\partial z} B_x(\vec{r}(t)) - \frac{\partial}{\partial x} B_z(\vec{r}(t)) \right] = 0, \quad (4.2.4)$$

where, of course, the two equations represent the divergence and the curl of the magnetic field.

One simplification which can be brought is to build the magnetic field profile such that it is invariant under reflection on the x - y plane:

$$\vec{B}(z(t)) = \vec{B}(-z(t)), \quad (4.2.5)$$

from which, using the two of the Maxwell's laws from above, it directly results that the following quantities must vanish at $\underline{z = 0}$ for any time t :

$$B_y(\vec{r}(t)) = \frac{\partial}{\partial z} B_y(\vec{r}(t)) = \frac{\partial}{\partial x} B_y(\vec{r}(t)) = \frac{\partial}{\partial y} B_z(\vec{r}(t)) = \frac{\partial}{\partial y} B_x(\vec{r}(t)) = 0. \quad (4.2.6)$$

There are two reasons behind the adoption of this symmetry: to simplify the field and its gradient for the sections to come and to obtain a symmetrical macroscopic split of the two states in the z direction, property heavily exploited in the following sections. All the following terms are now discussed at the same x - y plane at $z = 0$.

The next improvement is the vanishing of the z component of the field:

$$B_z(\vec{r}(t)) = \frac{\partial}{\partial z} B_z(\vec{r}(t)) = 0, \quad (4.2.7)$$

which, in fact, is true in the close proximity of the atom chips due to the fringing fields [2] and can thus be readily assumed for relatively small spatial displacements.

By making use once more of the above Maxwell's laws, the remaining components of the field are:

$$B_x(\vec{r}(t)) \quad \left| \quad \frac{\partial}{\partial y} B_y(\vec{r}(t)) = -\frac{\partial}{\partial x} B_x(\vec{r}(t)) \quad \left| \quad \frac{\partial}{\partial x} B_z(\vec{r}(t)) = \frac{\partial}{\partial z} B_x(\vec{r}(t)) = \frac{1}{v(t)} \frac{\partial}{\partial t} B_x(\vec{r}(t)), \quad (4.2.8)$$

where $v(t)$ is defined as the group velocity of the wave-packet in the z direction:

$$v(t) = \frac{1}{m} \langle p_z(t) \rangle \quad \rightarrow \quad v = \frac{1}{m} \langle p_z(t) \rangle, \quad (4.2.9)$$

which requires at least one non-zero entry in the third row of the field's Jacobian matrix (**Eq.4.2.2**), which now becomes (in the x - y plane at $\mathbf{z} = \mathbf{0}$ for any time $\mathbf{t}$):

$$\vec{\nabla} \vec{B}(\vec{r}(t)) = \begin{pmatrix} \frac{\partial}{\partial x} B_x(\vec{r}(t)) & 0 & \frac{\partial}{\partial z} B_x(\vec{r}(t)) \\ 0 & -\frac{\partial}{\partial x} B_x(\vec{r}(t)) & 0 \\ \frac{\partial}{\partial z} B_x(\vec{r}(t)) & 0 & 0 \end{pmatrix} = \begin{pmatrix} \frac{\partial}{\partial x} B_x(t) & 0 & \frac{1}{v} \frac{d}{dt} B_x(t) \\ 0 & -\frac{\partial}{\partial x} B_x(t) & 0 \\ \frac{1}{v} \frac{d}{dt} B_x(t) & 0 & 0 \end{pmatrix}. \quad (4.2.10)$$

The Jacobian is now simplifying to only two gradients of the x component of the field.

Given that the elements from each row represent the gradients in the three directions of each of the field's components respectively, one might argue the use of the velocity $v = v_z$ notation is sloppy, as the same component is found in the x direction (last row of the first column), besides the two other gradients on the main diagonal. The reason for this notation is that it is the only component that is not on the main diagonal (so it would be found in the divergence of the field) found alone on a column. How this bears any physical importance is detailed upon in the sections to come.

There is a very important observation to be made. If one further simplifies the field by reducing its Jacobian to:

$$\vec{\nabla} \vec{B}(\vec{r}(t)) = \begin{pmatrix} \frac{\partial}{\partial x} B_x(\vec{r}(t)) & 0 & 0 \\ 0 & -\frac{\partial}{\partial x} B_x(\vec{r}(t)) & 0 \\ 0 & 0 & 0 \end{pmatrix}, \quad (4.2.11)$$

which is allowed by Maxwell's equations, then, if only accounting for the macroscopic velocity of the states, their motion is reduced to the x - y plane.

Even more importantly, when accounting for their microscopical momentum spreads, their motion is constant in the z direction with constant acceleration in the other two, at any point in space. This is a very important observation, as it allows the replacement of the spatial dependence with that of time:

$$\vec{B}(\vec{r}(t)) = \begin{pmatrix} B_x(t) \\ 0 \\ 0 \end{pmatrix} \quad \left| \quad \vec{\nabla} \vec{B}(\vec{r}(t)) = \begin{pmatrix} \frac{\partial}{\partial x} B_x(t) & 0 & 0 \\ 0 & -\frac{\partial}{\partial x} B_x(t) & 0 \\ 0 & 0 & 0 \end{pmatrix}. \quad (4.2.12)$$

One aspect disregarded by the use of time dependence is that one state gains potential energy, while the other loses—thus, their time in the interferometer is also slightly different (for $\sigma_x = \pm 1$). This leads to a loss of spin coherence, unless their paths are not shortened or prolonged, respectively, with new field profiles, accordingly. Yet another overlooked point is that this difference in potential induce different probabilities of reflection at the boundaries of the SGI. The correction of those two observations is beyond the scope of this thesis, but their importance (especially that of the former) makes them at the very least worthy of mentioning.

4.3 Simplifications and Approximations

In the above section, a general magnetic field profile was created by setting two conditions: symmetry on the x - y plane and null magnetic field in the z direction on it.

In order for the procedure to be further simplified, the following three points are made, which also include the previous simplifications. The goal is to reduce the field only to its B_x component and its gradient to its $\frac{\partial}{\partial x} B_x$ one.

1. The fields are considered to be linear. - That is, there are no second-order spatially partial terms in the chosen profile. This approximation induces decoherence into the system, but, given how small the spatial displacement $\Delta \vec{r}$ is [13], the total effect is negligible, especially compared to the other sources of error.

2. B_y , B_z and their gradients (not present in **Eq.4.2.10) are very small and thus can be omitted.** - This simplification is, in essence the summary of the previous section in regards to the Jacobian matrix.

3. The particles have constant speed along the y and z axis. - This statement allows the changing the spatial dependence with a temporal one. Given the microscopic errors in the momentum $\vec{p}$ at time $t = 0$, the particle can have an original velocity, which, in order to be conserved, must pass through null gradients in those directions. The solution can be broken down by the following two equations:

$$\frac{\partial}{\partial y} B_y = 0 \quad \Bigg| \quad \frac{\partial}{\partial z} B_x = 0. \quad (4.3.1)$$

The latter can be set without any contradiction with any of Maxwell's equations but also renders the $\frac{\partial}{\partial x} B_z$ component null. The former, however, would nullify the most important gradient, $\frac{\partial}{\partial x} B_x$, which is not permitted in the SGI (would lead to all the gradient components being null). Thus, this approximation induces a considerable loss of coherence, which is touched upon in **Discussion**.

Like the change of the spatial dependence to that of time in the making of **Eq.4.2.12**, the macroscopic motion of the states is restricted to the x - y plane, but unlike in that case, the microscopic one is constant in all but the x direction, as it is the only one in which a field gradient acts upon.

One can summarise those last three simplifications as:

$$\vec{B}(\vec{r}(t)) \approx \begin{pmatrix} B_x(\vec{r}(t)) + \vec{r}^T(t) \cdot \vec{\nabla} B_y(\vec{r}(t)) \\ B_y(\vec{r}(t)) + \vec{r}^T(t) \cdot \vec{\nabla} B_y(\vec{r}(t)) \\ B_z(\vec{r}(t)) + \vec{r}^T(t) \cdot \vec{\nabla} B_z(\vec{r}(t)) \end{pmatrix} \approx \begin{pmatrix} B_x(\vec{r}(t)) + x(t) \frac{\partial}{\partial x} B_x(\vec{r}(t)) + z(t) \frac{\partial}{\partial z} B_x(\vec{r}(t)) \\ y(t) \frac{\partial}{\partial y} B_y(\vec{r}(t)) \\ 0 \end{pmatrix} \approx \begin{pmatrix} B_x(t) + x(t) \frac{\partial}{\partial x} B_x(t) \\ 0 \\ 0 \end{pmatrix}, \quad (4.3.2)$$

which represents all of the magnetic field's and its gradient's components. In mind of the Jacobian from **Eq.4.2.12**, the only approximation, in fact, the first part of the third one ($\frac{\partial}{\partial y} B_y(t) = 0$) as it violates Maxwell's second law.

In reality, the complete and correct form of the magnetic field vector is given by:

$$\vec{B}(\vec{r}(t)) = \vec{B}_h(\vec{r}(t)) + [\vec{\nabla}(\vec{r}(t) \cdot \vec{B}(\vec{r}(t)))] \cdot M_1 = \vec{B}_h(\vec{r}(t)) + \vec{\nabla}_M \vec{B}(\vec{r}(t)) \quad \Bigg| \quad M_1 = \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix} \quad \Bigg| \quad \vec{B}_h(\vec{r}(t)) = \begin{pmatrix} B_{h_x}(\vec{r}(t)) \\ B_{h_y}(\vec{r}(t)) \\ B_{h_z}(\vec{r}(t)) \end{pmatrix}, \quad (4.3.3)$$

where the subscript h denotes the homogenous component of the field (which is not found in the gradient) and $\vec{\nabla}_M$ is a newly introduced notation. This is cumbersome, but accurate. Thus, it should still be avoided where possible, but not dismissed. Thankfully, most of this thesis deals only with the x component of the field and in the few places where the general equations are provided, there are few places where the explicit formula above is needed.

Its purpose is to merely present a clear mathematical formulation of the field, even while at the price of cluttered notations. A compromise must be made, and, given how the context of each equation makes the distinction clear, the subscript h shall be dropped. However, the $\vec{\nabla}_M$ notation shall be kept.

Chapter 5 - Hamiltonian, Magnetic Fields and Operators

Before the formula for the coherence measure can be finally determined, a last round of "preparations" is in order. This chapter is dedicated to forming the equations of its constituents, making use of all the previous information. Thus, the main objectives are the determination of the hamiltonian, the annihilation and creation operators and making the shift from Schrödinger's picture to the Hermitian one.

5.1 Hamiltonian

In the view of the above sections, the following Hamiltonian is obtained:

$$H(t) = \frac{\vec{p}(t)^2}{2m} - \gamma \vec{\sigma} \cdot \vec{B}(\vec{r}(t)) \quad \Bigg| \quad H_x(t) = \frac{p_x(t)^2}{2m} - \gamma \sigma_x \left(B_x(t) - x(t) \frac{\partial}{\partial x} B_x(t) \right), \quad (5.1.1)$$

where, of course, the first and second terms in each of the right hand side (RHS) equations represent the kinetic and potential energy of the particle, respectively. The second equation was derived from the first using the simplifications from **Eq.4.3.2**.

Note that the Hamiltonian is time dependent. This is due to the sates probing the potential at different points as they move through the SGI and thus, its time dependence is in fact a spatial one, as it should, since it is inherited from the magnetic field term. This description pays off, as the Hamiltonian needs to be applied for every point on the wave-packet (which has a microscopical spread in position and momentum).

Of course, **Eq.5.1.1** is the term to which one has to go back and apply any simplification (the magnetic fields form **Chapter 4**), correction (accounting for factors such as fluctuations in the field and microscopic uncertainties) or even add new terms (such as the CP potential) to the potential experienced by the states.

5.2 Heisenberg Picture

In the above section, as discussed, the Hamiltonian's time dependence is but a hidden spatial one. Thus, it still lies in Schrödinger's picture. The change to the Heisenberg one is done through the use of the time-evolution creation ($\eta^\dagger$) and annihilation (η) operators as follows [18]:

$$\left. \begin{aligned} H_H(t) &= \eta^\dagger(t) H(t) \eta(t) \\ &= \eta^\dagger(t) \frac{\vec{p}(t)^2}{2m} \eta(t) - \eta^\dagger(t) \gamma \vec{\sigma} \cdot \vec{B}(\vec{r}(t)) \eta(t) \\ &= \frac{1}{2m} \vec{p}^2 - \gamma \eta^\dagger(t) \vec{\sigma} \cdot \vec{B}(\vec{r}(t)) \eta(t) \end{aligned} \right| \quad H_{Hx}(t) = \frac{1}{2m} p_x^2 - \gamma \eta^\dagger(t) \sigma_x \left(B_x(t) + x(t) \frac{\partial}{\partial x} B_x(t) \right) \eta(t), \quad (5.2.1)$$

where the kinetic energy term loss of time dependence was obtained using **Eq. 5.3.1** and the null state produced by the product of two evolution operators of the same type (**Chapter 5.4.1**). The complete derivation is found in **Appendix B**.

One should also note that the difference between the two pictures is the time-evolution as a whole, since time $t = 0$, the an operator in the Hermitian picture reduces itself to its counterpart from the Schrödinger one. This behavior is compatible to a (possible) complex exponential form of the two operators.

Moving on, the equation of motion of the annihilation operator is given by [19], [18]:

$$\left. \begin{aligned} \frac{d}{dt} \eta(t) &= i[H_H(t), \eta(t)] + \left(\frac{\partial \eta}{\partial t} \right)_H \\ &= i[\eta^\dagger(t) H(t) \eta(t), \eta(t)] + 0 \\ &= i \left[\frac{1}{2m} \vec{p}^2 - \gamma \eta^\dagger(t) \vec{\sigma} \cdot \vec{B}(\vec{r}(t)) \eta(t), \eta(t) \right] \\ &= i \left[\frac{1}{2m} \vec{p}^2, \eta(t) \right] - i\gamma [\eta^\dagger(t) \vec{\sigma} \cdot \vec{B}(\vec{r}(t)) \eta(t), \eta(t)] \\ &= 0 - i\gamma \left(\eta^\dagger(t) \vec{\sigma} \cdot \vec{B}(\vec{r}(t)) \eta(t) \eta(t) - \eta(t) \eta^\dagger(t) \vec{\sigma} \cdot \vec{B}(\vec{r}(t)) \eta(t) \right) \\ &= -i\gamma \left(0 - \vec{\sigma} \cdot \vec{B}(\vec{r}(t)) \eta(t) \right) \\ &= i\gamma \vec{\sigma} \cdot \vec{B}(\vec{r}(t)) \eta(t) \end{aligned} \right| \quad \frac{d}{dt} \eta_x(t) = i\gamma \sigma_x \left(B_x(t) + x(t) \frac{\partial}{\partial x} B_x(t) \right) \eta_x(t), \quad (5.2.2)$$

where $\left(\frac{\partial \eta_s}{\partial t} \right)_H$ represents the partial time derivative of the annihilation operator from Schrödinger's picture, which is then converted to the Hermitian one. Also, the annihilation of the $\eta(t)\eta(t)$ term from the fifth to sixth line of the equation on the left is observed, rendering the first term in the brackets null.

From this, it results directly that:

$$\left. \begin{aligned} \eta_x(t) &= \eta_x e^{i\gamma \sigma_x \int_0^t (B_x(t') + x(t') \frac{\partial}{\partial x} B_x(t')) dt'} \\ &= \eta_x e^{i\sigma_x A_x(t)} \\ &= \eta_x U_t(\sigma_x) \end{aligned} \right| \quad U_t(\sigma_x) = e^{i\sigma_x A_x(t)} \quad \left| \quad A_x(t) = \gamma \int_0^t \left(B_x(t') + x(t') \frac{\partial}{\partial x} B_x(t') \right) dt'. \quad (5.2.3)$$

These equations are the basis for all that is to follow, so one should indeed be very comfortable with them and their derivation. The equation of motion of the creation operator is obtained in an analogous manner and gives:

$$\frac{d}{dt} \eta^\dagger(t) = -i\gamma \vec{\sigma} \cdot \vec{B}(\vec{r}(t)) \eta^\dagger(t) \quad \left| \quad \frac{d}{dt} \eta_x^\dagger(t) = -i\gamma \sigma_x \left(B_x(t) + x(t) \frac{\partial}{\partial x} B_x(t) \right) \eta_x^\dagger(t) \quad \left| \quad \eta^\dagger(t) = \eta_x^\dagger U_t(-\sigma_x). \quad (5.2.4)$$

The U_t and A_x notations are used throughout the remaining of this thesis. While they depend on the field's profile, their form stays similar and thus all of the further equations still apply for the general case.

Making use of the fact that $U(\pm\sigma_x)$ is unitary:

$$U_t(\sigma_x)^\dagger = [U_t(\sigma_x)]^{-1} = U_t(-\sigma_x) \quad (5.2.5)$$

$$\eta_x^\dagger(t) = [\eta_x U_t(\sigma_x)]^\dagger. \quad (5.2.6)$$

Combining the first equations from **Eqs. 5.2.2** and **5.2.4**, by using the chain rule, one can immediately conclude:

$$\left. \begin{aligned} \frac{d}{dt}(\eta^\dagger(t)\eta(t)) &= \left(\frac{d}{dt}\eta^\dagger(t)\right)\eta(t) + \eta^\dagger(t)\left(\frac{d}{dt}\eta(t)\right) \\ &= -i\gamma\left(\eta^\dagger(t)\vec{\sigma} \cdot \vec{B}(\vec{r}(t))\eta(t) - \eta^\dagger(t)\vec{\sigma} \cdot \vec{B}(\vec{r}(t))\eta(t)\right) \\ &= 0 \end{aligned} \right| \quad \frac{d}{dt}(\eta_x^\dagger(t)\eta_x(t)) = 0. \quad (5.2.7)$$

But in the equations above, the term in the brackets on the LHS of each of them (which is thus constant) represents the total number of particles in the system—i.e., the number of $\frac{1}{2}$ -spin systems attributed to $\vec{S}$ and S_x , respectively, does not change with time.

The same process to obtain the Hermitian picture can be applied to the spin $\vec{S}$:

$$\vec{S}(t) = \eta^\dagger(t)\vec{S}\eta(t). \quad (5.2.8)$$

While each of the Pauli spin matrices are composed of complex numbers, one might rush to the conclusion that they commute with the creation and annihilation operators ($\eta^\dagger(t)$ and $\eta(t)$), which would result in a time-independent spin $\vec{S}$. This is bluntly obvious, from a physical standpoint—as discussed in **Chapter 3**—but what about the mathematics? In order to look for an answer, one must now expand the exponential into a sum:

$$U_t(\pm\sigma_x) = e^{\pm i\sigma_x A_x(t)} = \sum_{n=0}^{\infty} \frac{(\pm\sigma_x A_x(t))^n}{n!}, \quad (5.2.9)$$

from which, using the commutativity property of the Pauli spin matrices (specifically of σ_x), one obtains:

$$U_t(\pm\sigma_x)\sigma_x = \sum_{n=0}^{\infty} \frac{(\pm\sigma_x A_x(t))^n}{n!}\sigma_x = \sigma_x \sum_{n=0}^{\infty} \frac{(\pm\sigma_x A_x(t))^n}{n!} = \sigma_x U_t(\pm\sigma_x). \quad (5.2.10)$$

The following results directly:

$$\begin{aligned} S_x(t) &= \eta^\dagger(t)\frac{1}{2}\sigma_x\eta(t) = \eta^\dagger U_t(-\sigma_x)\frac{1}{2}\sigma_x\eta U_t(\sigma_x) = \frac{1}{2}\sigma_x(\eta^\dagger U_t(-\sigma_x))(\eta U_t(\sigma_x)) \\ &= \frac{1}{2}\sigma_x(\eta^\dagger\eta)(U_t(-\sigma_x)U_t(\sigma_x)) = \frac{1}{2}\sigma_x = S_x. \end{aligned} \quad (5.2.11)$$

But one can rewrite the above (and for the matter, redo the whole calculation) as the expectation value of the spin's x component as follows:

$$\langle S_x(t) \rangle = \langle S_x \rangle, \quad (5.2.12)$$

which translates into one of the most important statements in the case of the SGI: S_x stays constant in time. Thus, no matter the amount of decoherence, the expectation value of the x component of the spin remains constant.

This result is amazing! It is a confirmation of the simplified treatment from **Chapter 3.3**.

Accordingly, the same process cannot be done for the y and z components of the spin, as they do not commute with the x one from the exponentials in the $U_t(\pm\sigma_x)$ terms. Of course, the origin of this property comes from the form of the evolution operators which, at its turn, originates from the magnetic field profile selected, whose only components (and gradients) are in the x direction.

5.3 General Equations of Motion

As seen in the above section (notably in **Eq.5.2.7**), it pays off to first sometimes also determine the general form of the equations encountered. Of course, this also leaves some "headroom" for a future more complete description of the fields. Thus, one ought to start with the basic equations correlating:

- momentum to the magnetic fields and spins
- position to momentum,

which can be written as such:

$$\frac{d}{dt}\vec{p}(t) = 2\gamma \left[\vec{\nabla} \vec{B}(\vec{r}(t)) \circ \vec{S}(t) \right] \quad \mapsto \quad \vec{p}(t) = \vec{p} + \gamma \int_0^t \vec{\nabla} \vec{B}(\vec{r}(t')) \circ (\eta^\dagger(t') \vec{\sigma} \eta(t')) dt' \quad (5.3.1)$$

$$\begin{aligned} \frac{d}{dt}\vec{r}(t) = \frac{1}{m}\vec{p}(t) \quad \mapsto \quad \vec{r}(t) &= \vec{r} + \frac{t}{m}\vec{p} + \frac{\gamma}{m} \int_0^t (t-t') \vec{\nabla} \vec{B}(\vec{r}(t')) \circ (\eta^\dagger(t') \vec{\sigma} \eta(t')) dt' \\ &= \vec{r} + \frac{t}{m}\vec{p}(t) + \frac{\gamma}{m} \int_0^t (-t') \vec{\nabla} \vec{B}(\vec{r}(t')) \circ (\eta^\dagger(t') \vec{\sigma} \eta(t')) dt'. \end{aligned} \quad (5.3.2)$$

One notes that the, since the magnetic field's components are scalars, they commute with the evolution operators, which allowed its placement in the integrals on the left of the Hermitian spin. There is, in fact a very good reason for this exact positioning, which is exploited in the beginning of the following section.

A definition the displacements of position and momentum is now in order and, while some of them have already been mentioned in the previous chapters, it is good practice to summarize all of them in one place, especially given the newly adopted notation for the field's arguments. They follow directly from the above two equations:

$$\Delta\vec{p}(t) = \gamma \int_0^t \vec{\nabla} \cdot \vec{B}(\vec{r}(t')) dt' \quad \left| \quad \begin{aligned} \Delta\vec{r}(t) &= \frac{\gamma}{m} \int_0^t (t-t') \vec{\nabla} \cdot \vec{B}(\vec{r}(t')) dt' \\ \bar{\Delta}\vec{r}(t) &= \Delta\vec{r}(t) - \frac{t}{m} \Delta\vec{p}(t) = \frac{\gamma}{m} \int_0^t (-t') \vec{\nabla} \cdot \vec{B}(\vec{r}(t')) dt', \end{aligned} \right. \quad (5.3.3)$$

where $\Delta\vec{r}(t)$ and $\Delta\vec{p}(t)$ represent the displacements of the partial beams in position and momentum. Also, the vectorial counterpart of **Eq.3.2.4** of the Larmor angle is given by:

$$\vec{\Phi}(t) = 2\gamma \int_0^t \vec{B}(\vec{r}(t')) dt'. \quad (5.3.4)$$

5.4 Magnetic Field: Macro and Micro

5.4.1 Null Spin State

For the $\frac{1}{2}$ -spin state, the vector operators $\langle\eta^\dagger(t)$ and $\eta(t)\rangle$ can easily be built using **Eq.5.2.3** and **Eq.5.2.4**, by simply adding the *up* and *down* states of the spin before and after them, respectively.

By making use of them and rewriting the argument of the magnetic field $\vec{B}(\vec{r}(t))$ using **Eq.5.3.2**, one obtains the following:

$$\begin{aligned}\vec{B}(\vec{r}(t))\eta(t)\rangle &= \vec{B}\left(\vec{r} + \frac{t}{m}\vec{p} + \int_0^t (t-t')\frac{\gamma}{m}\vec{\nabla}\vec{B}(\vec{r}(t')) \circ (\eta^\dagger(t')\vec{\sigma}\eta(t'))dt'\right)\eta(t)\rangle \\ &= \vec{B}\left(\vec{r} + \frac{t}{m}\vec{p}\right)\eta(t)\rangle.\end{aligned}\tag{5.4.1}$$

This is strange. How was the integral term in the field's argument reduced? One might observe the consecutive annihilation operators and argue they just cancel out, but this would not explain how they passed through the integral and into the argument to interact.

The *up* state is multiplied by the annihilation operator, thus resulting in a null spin state [20]. Combined, the first two components in the argument of the magnetic field (as in the first line of the above equation) represent the position of the particle at time t on which the field acts on based only on its original position add momentum.

The third term, however, is given by the time integral of the evolution of momentum integral (the equation of the left in **Eq.5.3.3**). Thus, nullifying this term represents a constant momentum of the states. That is, there is no spin of the states to provide an increase in momentum due to its interaction with the magnetic field.

An analogous argument can be made for the annihilation operator, which leads to:

$$\langle\eta^\dagger(t)\vec{B}(\vec{r}(t)) = \langle\eta^\dagger(t)\vec{B}\left(\vec{r} + \frac{t}{m}\vec{p}\right).\tag{5.4.2}$$

This is the reason for the specific choice of positioning made at the end of the previous section.

As a last point, one can finally see the advantage of writing the spin $\vec{S}(t)$ using the annihilation-creation description of the $\frac{1}{2}$ -spin state (**Eq.5.2.8**).

5.4.2 Macro and Micro

Both the microscopic and macroscopic scales of position and momentum must be accounted for. Given that the problem is now set in the states' reference frames, a more sophisticated magnetic field profile is in order. To achieve it, the expectation values of the position and momentum are used to better illustrate the difference between the two scales, by rewriting the following quantity:

$$\vec{r}(t) + \frac{t}{m}\vec{p}(t) = \left(\langle\vec{r}(t_1)\rangle + \frac{t}{m}\langle\vec{p}(t_1)\rangle\right) + \left[\left(\vec{r}(t) - \langle\vec{r}(t_1)\rangle\right) + \frac{t}{m}(\vec{p}(t) - \langle\vec{p}(t_1)\rangle)\right].\tag{5.4.3}$$

This form separates the position and momentum as a sum of their expectation values at a general time t_1 and their difference to the same expectation values (at the same t_1). In particular, for equal times at the entrance in the SGI ($t = t_1 = 0$), the above equation becomes:

$$\vec{r} + \frac{t}{m}\vec{p} = \left(\langle\vec{r}\rangle + \frac{t}{m}\langle\vec{p}\rangle\right) + \left[\left(\vec{r} - \langle\vec{r}\rangle\right) + \frac{t}{m}(\vec{p} - \langle\vec{p}\rangle)\right].\tag{5.4.4}$$

This allows the magnetic field from **Eq.5.4.1** and **Eq.5.4.2** to be rewritten as follows:

$$\begin{aligned}\vec{B}\left(\vec{r} + \frac{t}{m}\vec{p}\right) &= \vec{B}\left(\langle\vec{r}\rangle + \frac{t}{m}\langle\vec{p}\rangle + \left[(\vec{r} - \langle\vec{r}\rangle) + \frac{t}{m}(\vec{p} - \langle\vec{p}\rangle)\right]\right) \\ &\approx \vec{B}\left(\langle\vec{r}\rangle + \frac{t}{m}\langle\vec{p}\rangle\right) + \left((\vec{r} - \langle\vec{r}\rangle) + \frac{t}{m}(\vec{p} - \langle\vec{p}\rangle)\right) \circ \vec{\nabla} \vec{B}\left(\langle\vec{r}\rangle + \frac{t}{m}\langle\vec{p}\rangle\right) \\ &= \vec{B}(t) + \left((\vec{r} - \langle\vec{r}\rangle) + \frac{t}{m}(\vec{p} - \langle\vec{p}\rangle)\right) \circ \vec{\nabla} \vec{B}(t)\end{aligned}\tag{5.4.5}$$

$$B_x\left(\vec{r} + \frac{t}{m}\vec{p}\right) \approx B_x(t) + \left[(x - \langle x \rangle) + \frac{t}{m}(p_x - \langle p_x \rangle)\right] \frac{\partial}{\partial x} B_x(t).\tag{5.4.6}$$

In the second line of the general equation (**Eq.5.4.5**) an approximation has been made. It linearized the field into a component that only depends on the expectation values of position and momentum—representing the averages of the wave-packets at time t —and its gradient. The later is multiplied by a term denoting the spread of the wave-packets. Thus, the equation above accentuates the quantum nature of the SGI, unlike **Eq.4.3.2**, which it is reminiscent of. Of course, since the only variable left in the argument was that time t , it became the only term in the argument:

$$\langle\vec{r}\rangle + \frac{t}{m}\langle\vec{p}\rangle \rightarrow t.\tag{5.4.7}$$

Note, however, that $\vec{B}(t)$ still does not represent a time-varyingly field. There was no such (fundamental) change brought to it.

A very important point to be made here, which becomes critical in the following section, is that in going from the forth to fifth line of **Eq.5.4.5**, only the position and momentum terms from the bracket in front of the field's gradient have been replaced with their x components and not those from the arguments of the field and its gradient. This is because those two represent the "initial setup" of the field and, of course, originate from the three dimensional space, even if the problem is restricted to just one direction (the x axis):

$$B_x(t) = B_x\left(\langle\vec{r}\rangle + \frac{t}{m}\langle\vec{p}\rangle\right) \quad \left| \quad \frac{\partial}{\partial x} B_x(t) = \frac{\partial}{\partial x} B_x\left(\langle\vec{r}\rangle + \frac{t}{m}\langle\vec{p}\rangle\right).\tag{5.4.8}$$

5.4.3 Reduced Equations of Motion

Making use of the simplifications made in the above sections and restricting the magnetic field and its gradient to the x direction and of **Eq.5.2.11**, the general equations of motion from **Chapter 5.3** are reduced to:

$$\begin{aligned}p_x(t) &= p_x + \gamma \int_0^t \frac{\partial}{\partial x} B_x(t') \cdot (\eta^\dagger(t') \sigma_x \eta(t')) dt' \\ &= p_x + \gamma \int_0^t \frac{\partial}{\partial x} B_x(t') \cdot \sigma_x dt' \\ &= p_x + \gamma \sigma_x \int_0^t \frac{\partial}{\partial x} B_x(t') dt' \\ &= p_x + 2S_x \Delta p_x(t)\end{aligned} \quad \left| \quad \begin{aligned}x(t) &= x + \frac{t}{m} p_x + \frac{\gamma}{m} \int_0^t (t - t') \frac{d}{dx} B_x(t') \cdot (\eta^\dagger(t') \sigma_x \eta(t')) dt' \\ &= x + \frac{t}{m} p_x + \frac{\gamma}{m} \int_0^t (t - t') \frac{d}{dx} B_x(t') \cdot \sigma_x dt' \\ &= x + \frac{t}{m} p_x + \frac{\gamma}{m} \sigma_x \int_0^t (t - t') \frac{d}{dx} B_x(t') dt' \\ &= x + \frac{t}{m} p_x + 2S_x \Delta x(t)\end{aligned}\tag{5.4.9}$$

$$\Delta p_x(t) = \gamma \int_0^t \frac{\partial}{\partial x} B_x(t') dt' \quad \left| \quad \begin{aligned} \Delta x(t) &= \frac{\gamma}{m} \int_0^t (t-t') \frac{\partial}{\partial x} B_x(t') dt' \\ \bar{\Delta}x(t) &= \Delta x(t) - \frac{t}{m} \Delta p_x(t) = \frac{\gamma}{m} \int_0^t (-t') \frac{\partial}{\partial x} B_x(t') dt'. \end{aligned} \right. \quad (5.4.10)$$

Combining the two sides of **Eq.5.4.9** with the formula for $\bar{\Delta}x(t)$ from the above equation, one obtains:

$$x(t) - x - \frac{t}{m} p_x(t) = 2S_x \cdot \bar{\Delta}x(t) \quad \left| \quad \Phi_x(t) = 2\gamma \int_0^t B_x(t') dt', \quad (5.4.11)$$

where the second equation represents the Larmor angle given by the x component of the spin.

Of course, the HD problem requires accountability of all the potential sources in the Hamiltonian. Therefore, in the case of the QGEM protocol, potentials such as the CP and the gravitational ones modify the above equations.

Chapter 6 - Spin Coherence

All the preparations required have been made: it is time to tackle the primary goal of this thesis—obtaining a formula for the coherence measure C of a $\frac{1}{2}$ -spin particle in an SGI.

6.1 Time-Evolution Operators

By interesting **Eq.5.4.6** into **Eq.5.2.2**, the later can be rewritten as:

$$\frac{d}{dt}\eta_x(t) = i\gamma\sigma_x \left\{ B_x(t) + \left[(x - \langle x \rangle) + \frac{t}{m}(p_x - \langle p_x \rangle) \right] \frac{\partial}{\partial x} B_x(t) \right\} \eta_x(t). \quad (6.1.1)$$

If one further assumes that that the particles were prepared such that $\langle p_x \rangle = \langle x \rangle = 0$ (a very hard task in practice, but nonetheless achievable to a very good degree of approximation), the following is obtained:

$$\frac{d}{dt}\eta_x(t) = i\gamma\sigma_x \left[B_x(t) + \left(x + \frac{t}{m}p_x \right) \frac{\partial}{\partial x} B_x(t) \right] \eta_x(t). \quad (6.1.2)$$

Thus, one can apply the same procedure as in **Eq.5.2.4**, the final expression of the time dependent annihilation operator $\eta_x(t)$ is given by (the complete calculation is found in **Appendix C**):

$$\eta_x(t) = U_t(\sigma_x)\eta_x \quad \text{for} \quad U_t(\sigma_x) = e^{i\sigma_x \frac{1}{2}\Phi_x(t)} e^{i\sigma_x (x\Delta p_x(t) - p_x\bar{\Delta}x(t))}. \quad (6.1.3)$$

In retrospective, one can now see the full form of the $A(t)$ term from **Eq.5.2.3** is given by:

$$A_x(t) = \sigma_x \left[\frac{1}{2}\Phi_x + (x\Delta p_x(t) - p_x\bar{\Delta}x(t)) \right]. \quad (6.1.4)$$

6.2 S_+ - Expectation Values

From this point onwards, following the arguments from **Chapter 3.3**, the detailed study of the S_+ ladder operator's evolution becomes feasible and thus becomes the main focus.

In analogue manner to the determination of **Eq.5.2.11**, its expectation value is given by:

$$\langle S_+(t) \rangle = \langle (S_y(t) + iS_z(t)) \rangle = \langle \eta^\dagger(T) \frac{1}{2} (\sigma_y + i\sigma_z) \eta(T) \rangle = \langle \eta^\dagger U_T(-\sigma_x) \frac{1}{2} (\sigma_x + i\sigma_z) U_T(\sigma_x) \eta \rangle. \quad (6.2.1)$$

Keeping in mind the expansion of the complex exponential $U_t(\sigma_x)$ from **Eq.5.2.9**, the first product from the left ($S_+ U_t(\sigma_x)$) causes the terms in its argument to be multiplied by the ladder operator S_+ , thus replacing them with:

$$\sigma_+ \cdot \sigma_x = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix} \cdot \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} = \begin{pmatrix} 0 & -1 \\ 0 & 0 \end{pmatrix} = -S_+, \quad (6.2.2)$$

where only the -1 entry of σ_x was utilized. Next, the product between $U_t(-\sigma_x)$ and the result of the above equation is given by:

$$(-\sigma_x) \cdot (-S_+) = \sigma_x \cdot S_+ = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix} = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix} = S_+, \quad (6.2.3)$$

where only the 1 entry of σ_x was utilized.

Thus, one can replace the RHS and LHS σ_x 's with ∓ 1 terms and leave the middle term the same (as it is technically replaced by the result of **Eq.6.2.3**):

$$\langle S_+(t) \rangle = \langle \eta^\dagger \frac{1}{2} (\sigma_y + i\sigma_z) \eta [U_t(-1)]^2 \rangle = \langle S_+ \rangle \langle [U_t(-1)]^2 \rangle. \quad (6.2.4)$$

The brilliance of this result lies in the fact that the ∓ 1 terms represent actual numbers and not the identity matrix, which means that the $U_t(\mp 1)$ terms can now commute with the spin term in the middle and thus be taken separately. Put in physical terms, all the remaining dependence on spin is now contained in the $\langle S_+ \rangle$ term.

With **Chapter 3.3** in mind once again, one is now aiming directly for the ratio of ladder operator's expectation values at the beginning and end of the SGI:

$$\frac{\langle S_+(T) \rangle}{\langle S_+ \rangle} = \langle [U_T(-1)]^2 \rangle = e^{-i\Phi(T)} \langle e^{-2i(x\Delta p_x(T) - p_x \bar{\Delta}x(T))} \rangle = e^{-i\Phi(T)} C, \quad (6.2.5)$$

where C is the long sought after coherence measure, finally appearing in a proper mathematical form:

$$C = \langle e^{-2i(x\Delta p_x(T) - p_x \bar{\Delta}x(T))} \rangle. \quad (6.2.6)$$

One recognizes the complex exponential in front of C (containing the Larmor angle at time T) as the expectation value of the ladder operator in the absence of any decoherence (as discussed in **Chapter 3.3**).

A mention is in order: of course, C has an explicit time dependence; however, it is not put in the argument or even in the subscript because the only time of interest in this thesis is the final $t = T$ one. Other terms such as $U_T(\sigma_x)$ do have this dependence explicitly stated because, for more general treatments such as the inclusion of multiple field profiles, they need to be split into multiple components representative of different time intervals.

6.3 Stationary States: Phase Addition

In order to further advance in the endeavor of finding the explicit form of C , one needs to start by replacing the *up* and *down* states with their integral forms:

$$C = \int_{-\infty}^{\infty} \Psi_0(x)^* e^{-2i(x\Delta p_x(T) - p_x \bar{\Delta}x(T))} \Psi_0(x) dx = \int_{-\infty}^{\infty} \Psi_0(p_x)^* e^{-2i(x\Delta p_x(T) - p_x \bar{\Delta}x(T))} \Psi_0(p_x) dp_x, \quad (6.3.1)$$

where the first integration is that of the coordinate wave function and the later if its respective momentum one. Despite both forms leading to the same form of C , they are considered separately. The reason is that, even while the general steps are the same between the two, the complications to be overcome are different in the transition from $\bar{\Delta}x(T)$ to $\Delta x(T)$. The general notation for the momentum-space wave function is Φ , but, in order to avoid confusion with the Larmor angle, it shall be replaced with Ψ . The difference between the two is visible in their arguments.

After the particle is left out of the trap and enters the SGI, it is composed of a BEC and thus can be attributed a single wave function [7].

The following question arises: what is the form of $\Psi_0(x)$? One might think that it is Schrödinger's Gaussian (**Eq.2.3.1**), but that only comes later into play.

Schrödinger's Gaussian represents a wave-packet. It comprises an infinity of stationary states (free particles) with different momenta/positions, following a specific distribution, respectively. It is those states that need to be initially plotted into **Eq.6.3.1** and only then be integrated over distributions to form the packet. If this seems strange and perhaps even unnecessary—why shouldn't one start right away with the Gaussian—perhaps a closer look at the form of the exponential in **Eq.6.3.1** is in order.

It has two components, each having a displacement term multiplied by either the position and momentum at time $t = 0$, respectively. Those complex exponential factors represent the accumulated phases induced by the SGI. Thus, they act directly on the constituents of the wave-packet: the stationary states. It is by this mechanism that the phase is introduced in the packet and not left "outside" of it with no accountability. Therefore, one ought to start with the wave function of a stationary state and its conjugate, imbed the different phases into them, and only then introduce their respective momentum or position distributions to start their integration process. That is, one should not begin with the integrals found in **Eq.6.3.1**, which only come afterward. The result consists of either position or momentum-independent wave-packets with their respective conjugates and the remaining complex exponential between them, which can then be together integrated into the final expression of C . A counter-example is provided in **Appendix G**.

Of course, the changes brought to the wave-packet are assumed to be adiabatic and therefore to not change its shape, maintaining its position and momentum distributions—that is, obviously, besides the phase shift.

The general coordinate wave function of a standing wave is given by:

$$\Psi_t(x) = A e^{i(xp_x - Et)}, \quad (6.3.2)$$

where A is a normalization constant, E is the energy of the state and x is the eigenvalue of the operator x , thus the "coordinate" part in the naming—the same applies to p_x . While the notation might seem confusing, if used and stated accordingly, its consistent use is clear and should not cause issues.

Of course, the above wave function (**Eq.6.3.2**) completely cancels out with its conjugate. Since there are no matrix terms nor operators in a position to prevent any commutation inside of **Eq.6.3.1**, it can also do so inside of it. However, this would go directly against the above point: not only would this integration process lead to a complete miss of the phase addition, but it can even result in infinities (for $p_x = 0$).

Schrödinger's Gaussian (**Eq.2.3.1**) is obtained by integrating **Eq.6.3.2** over the entire momentum space:

$$\Psi_t^G(x) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} \Phi_t^G(p_x) e^{ixp_x} dp_x, \quad (6.3.3)$$

where $\Phi_t^G(p_x)$ is the momentum-space wave function—which was earlier referred as the momentum distribution of the standing waves forming the Gaussian packet. One can make use of either Fourier's Trick or Plancherel's theorem [21] to obtain:

$$\Psi_t^G(p_x) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} \Psi_t^G(p_x) e^{-ixp_x} dp_x. \quad (6.3.4)$$

From the above, one observes that the momentum-space analogue of **Eq.6.3.2** is given by the extra term in the above integral with the attached time dependence given by the particle's energy:

$$\Psi_t(p_x) = B e^{-i(xp_x - Et)}. \quad (6.3.5)$$

Making use of **Eqs.6.3.2** and **6.3.5**, one can perform the phase addition in the two forms of **Eq.6.3.1** respectively, and obtain (see **Appendix D** for the full calculation):

$$C = \int_{-\infty}^{\infty} \Psi_0(x - \bar{\Delta}x(T))^* e^{-2ix\Delta p_x(T)} \Psi_0(x + \bar{\Delta}x(T)) dx = \int_{-\infty}^{\infty} \Psi_0(p_x - \Delta p_x(T))^* e^{2ip_x\bar{\Delta}x(T)} \Psi_0(p_x + \Delta p_x(T)) dp_x. \quad (6.3.6)$$

One now observes how the momentum-space wave function leads to a similar form of **Eq.2.3.1**:

$$\Psi_0^G(p_x) = (2\pi)^{-\frac{1}{4}} (\delta p_x)^{-\frac{1}{2}} e^{i\langle x \rangle (p_x - \langle p_x \rangle)} e^{-\frac{1}{4} \left(\frac{p_x - \langle p_x \rangle}{\delta p_x} \right)^2}. \quad (6.3.7)$$

The objective becomes to replace $\bar{\Delta}x(T)$ with $\Delta x(T)$. One starts now to look at the time dependence of the stationary states (where, obviously, the same argument goes for both $\Psi_t(x)$ and $\Phi_t(p_x)$, leading to the same time dependence):

$$\Psi_t(x) = e^{-iEt} \Psi_0(x) = e^{-ip_x^2 \frac{t}{2m}} \Psi_0(x) \quad \Bigg| \quad \Psi_t(p_x) = e^{-iEt} \Psi_0(p_x) = e^{-ip_x^2 \frac{t}{2m}} \Psi_0(p_x), \quad (6.3.8)$$

leading directly to:

$$\begin{aligned} \frac{\Psi_t(x - \bar{\Delta}x(T))^* \Psi_t(x - \bar{\Delta}x(T))}{\Psi_0(x - \bar{\Delta}x(T))^* \Psi_0(x - \bar{\Delta}x(T))} &= \frac{\Psi_t(p_x - \Delta p_x(T))^* \Psi_t(p_x - \Delta p_x(T))}{\Psi_0(p_x - \Delta p_x(T))^* \Psi_0(p_x - \Delta p_x(T))} = \left(e^{-i(p_x - \Delta p_x(T))^2 \frac{t}{2m}} \right)^* e^{-i(p_x + \Delta p_x(T))^2 \frac{t}{2m}} \\ &= \left(e^{i(p_x - \Delta p_x(T))^2 \frac{t}{2m}} \right) e^{-i(p_x + \Delta p_x(T))^2 \frac{t}{2m}} \\ &= e^{-i2p_x \Delta p_x(T) \frac{t}{m}}. \end{aligned} \quad (6.3.9)$$

Applying the above equation to **Eq.6.3.6** gives:

$$\begin{aligned} C &= \int_{-\infty}^{\infty} \Psi_T(x - \bar{\Delta}x(T))^* e^{-2ix\Delta p_x(T)} \Psi_T(x + \bar{\Delta}x(T)) e^{2ip_x\Delta p_x(T) \frac{T}{m}} dx \\ &= \int_{-\infty}^{\infty} \Psi_T(x - \bar{\Delta}x(T))^* e^{-2ix\Delta p_x(T)} \Psi_T(x + \bar{\Delta}x(T)) e^{2ip_x[\Delta x(T) - \bar{\Delta}x(T)] \frac{T}{m}} dx \\ &= \int_{-\infty}^{\infty} \Psi_T(x - \Delta x(T))^* e^{-2ix\Delta p_x(T)} \Psi_T(x + \Delta x(T)) dx \end{aligned} \quad (6.3.10)$$

$$\begin{aligned} C &= \int_{-\infty}^{\infty} \Psi_T(p_x - \Delta p_x(T))^* e^{2ip_x\bar{\Delta}x(T)} \Psi_T(p_x + \Delta p_x(T)) e^{2ip_x\Delta p_x(T) \frac{T}{m}} dp_x \\ &= \int_{-\infty}^{\infty} \Psi_T(p_x - \Delta p_x(T))^* e^{2ip_x[\Delta x(T) + (-\Delta p_x(T) + \Delta p_x(T)) \frac{T}{m}]} \Psi_T(p_x + \Delta p_x(T)) dp_x \\ &= \int_{-\infty}^{\infty} \Psi_T(p_x - \Delta p_x(T))^* e^{2ip_x\Delta x(T)} \Psi_T(p_x + \Delta p_x(T)) dp_x, \end{aligned} \quad (6.3.11)$$

where in the transitions from the first to the second lines of both the equations above were made using the third part of **Eq.5.4.10**. One can now appreciate how the two routes behave differently, yet are always in tandem, leading to the same general forms (only differing by a minus in the exponentials to account for the difference between the position and momentum spaces, as illustrated by Fourier's trick).

6.4 Schrödinger's Gaussian: Coherence Measure

6.4.1 Finding C

All the preparations have been made: it is time to use Schrödinger's Gaussian.

Given **Eqs.6.3.10** and **6.3.11**, one sees how the time dependence of the Gaussians is lost in the integration process due to the complex exponentials they represent, the first one being conjugated—any common complex phase they have cancels-out when searching for the wave-packets' overlap. Thus, **Eqs.2.3.1** and **6.3.7** can be reduced to:

$$\Psi_0^G(x) = (2\pi)^{-\frac{1}{4}}(\delta x)^{-\frac{1}{2}}e^{-\frac{1}{4}\left(\frac{x}{\delta x}\right)^2} \quad \Bigg| \quad \Psi_0^G(p_x) = (2\pi)^{-\frac{1}{4}}(\delta p_x)^{-\frac{1}{2}}e^{-\frac{1}{4}\left(\frac{p_x}{\delta p_x}\right)^2}. \quad (6.4.1)$$

An important observation about the complex exponential phases from the formulas of C is that they are not symmetrical around $x = 0$: they have both a real and a complex component (each) given by the (even) cosine and (odd) sine, respectively, of their arguments. Thus, for a general wave-packet, this causes the C to have both a real and complex component. However, the Gaussian wave-packets are symmetrical around $x = 0$:

$$\Psi_t^G(x) = \Psi_t^G(-x) \quad \Bigg| \quad \Psi_t^G(p_x) = \Psi_t^G(-p_x), \quad (6.4.2)$$

which leave only the sine component to produce an anti-symmetric integration, which, given its limits ($\pm\infty$), reduces itself to zero. Thus, one sees that such a wave-packet produces a real value of the coherence measure ($C \in \mathbb{R}$).

There is one more advantage to the above equalities: they directly imply (while they are a much stronger argument than necessary) null expectation values of both position and momentum:

$$\langle x \rangle = \langle p_x \rangle = 0. \quad (6.4.3)$$

This is one of the main advantages of working with Schrödinger's Gaussian and, for the interpretation of C from this thesis, this symmetry is required.

Therefore, based on the overlap of the two wave functions in the last lines of **Eqs.6.3.10** and **6.3.11**, for a good result, the splitting (SG) and recombination (SGI) of the beams must result in:

- **SG** $\rightarrow |C| \approx 0$ — The coherence in the y - z plane is destroyed due to no overlap ($\Delta x \gg \delta x$ and $\Delta p_x \gg \delta p_x$).
- **SGI** $\rightarrow |C| \approx 1$ — The coherence in the y - z plane is restored due to an almost perfect overlap ($\Delta x \sim \delta x$ and $\Delta p_x \sim \delta p_x$).

One should note that the above conditions do not put any restriction on the time spent in the $[T_2, T_3]$ stage of the experiment (see **Fig.2.1**), and it is time when the homogenous magnetic field can fine-tune the Larmor angle in the SGI.

By inserting the two forms of the Gaussian wave-packet (**Eq.6.4.1**) into the the last lines of **Eq.6.3.10** and **Eq.6.3.11**, respectively, one obtains (see **Appendix E** for the complete derivation):

$$C = (2\pi)^{-\frac{1}{2}}(\delta x)^{-1}e^{-\frac{1}{2}\left(\frac{\Delta x(T)}{\delta x}\right)^2} \int_{-\infty}^{\infty} e^{-\frac{1}{2(\delta x)^2}x^2} e^{i(-2\Delta p_x(T))x} dx. \quad (6.4.4)$$

$$C = (2\pi)^{-\frac{1}{2}}(\delta p_x)^{-1}e^{-\frac{1}{2}\left(\frac{\Delta p_x(T)}{\delta p_x}\right)^2} \int_{-\infty}^{\infty} e^{-\frac{1}{2(\delta p_x)^2}p_x^2} e^{i(2\Delta x(T))p_x} dp_x \quad (6.4.5)$$

One notes that, if the momentum and respectively position phase addition from **Eq.6.3.6** would have occurred only in one of the two Gaussians present in the integral—which, from a pure mathematical standpoint, is allowed—the x^2 and p_x^2 terms from the above integrals would be replaced by $x^2 \pm 2x\Delta x(T)$ and $p_x^2 \pm 2p_x\Delta p_x(T)$, respectively, where the plus term depends on the chosen Gaussian. Performing this phase addition, however, would completely disregard that, inside the SGI, the states are both undergoing the separation and recombination process, not just one of the two. It is in this stage, obviously, when the partition of the accumulated displacements is partitioned between the two packets.

The product of the overlap in **Eqs.6.4.5** and **6.4.5** is, as expected, yet another Gaussian. The exponential term—which, in view of the above discussion, reduces itself down to a cosine term—induces an oscillatory behavior inside the packet. Unless $\Delta x(T) = \Delta p_x(T) = 0$ —the microscopic criteria which makes the HD so hard to be put back together—it is not a "normalized packet" anymore. In fact, the term "packet" is misleading since one is only interested in its integral while in this form and not that of its squared modulus. Thus, using sloppy language, C tests "how normalized" the newly formed "packet" is.

The above integrals are recognized to have the following general shape:

$$\int_{-\infty}^{\infty} e^{-ax^2} e^{ibx} dx = e^{-\frac{b^2}{4a}} \int_{-\infty}^{\infty} e^{-a(x - \frac{ib}{2\sqrt{a}})^2} dx = e^{-\frac{b^2}{4a}} \int_{-\infty}^{\infty} e^{-\frac{(x - \frac{ib}{2\sqrt{a}})^2}{(\frac{1}{\sqrt{a}})^2}} dx = e^{-\frac{b^2}{4a}} \sqrt{\pi} \left| \sqrt{\frac{1}{a}} \right|, \quad (6.4.6)$$

which can also be recognized as a more general form of the Hubbard-Stratonovich transformation [22].

Making use of the above formula, one can rewrite **Eq.6.4.5** (with **Eq.6.4.4** following in analogous manner) as follows:

$$\begin{aligned} a = \frac{1}{2(\delta p_x)^2} \quad | \quad b = -2\Delta x(T) \longrightarrow C &= \left[(2\pi)^{-\frac{1}{2}} (\delta p_x)^{-1} e^{-\frac{1}{2} \left(\frac{\Delta p_x(T)}{\delta p_x} \right)^2} \right] \left[e^{-\frac{(-2\Delta x(T))^2}{4 \frac{1}{2(\delta p_x)^2}}} \sqrt{\pi} \left| \sqrt{2\delta p_x} \right| \right] \\ &= e^{-\frac{1}{2} \left(\frac{\Delta p_x(T)}{\delta p_x} \right)^2 - 2(\Delta x(T) \delta p_x)^2} = e^{-\frac{1}{2} \left(\frac{\Delta p_x(T)}{\delta p_x} \right)^2 - 2 \left(\Delta x(T) \frac{1}{2\delta x} \right)^2} \\ &= e^{-\frac{1}{2} \left[\left(\frac{\Delta p_x(T)}{\delta p_x} \right)^2 + \left(\frac{\Delta x(T)}{\delta x} \right)^2 \right]}. \end{aligned} \quad (6.4.7)$$

This is it. This is the final formula for the coherence measure C and, relievingly, it perfectly matches that from J. Schwinger's papers [1], [2].

6.4.2 The Perfect SGI

Idealistically, in one's attempt to obtain the best results possible in the QGEM protocol, the the two states must be perfectly recombined ($C = 1$). Looking at the last line of **Eq.6.4.7**, this requirement translates to:

$$\Delta x(T) = 0 \ll \delta x \quad | \quad \Delta p_x(T) = 0 \ll \delta p_x. \quad (6.4.8)$$

This is nothing new: the precision required must be sub-microscopical. However, there is one more variable which must be optimized—the Larmor angle $\Phi(T)$. In **Chapter 3.3**, the expectation value of the spin's components at times $t = 0$ and $t = T$ were set as follows (**Eq.3.3.1**):

$$\langle 2S_y \rangle = 1 \quad | \quad \langle 2S_z \rangle = 0 \quad | \quad \langle 2S_x \rangle = 0$$

$$\langle 2S_y(t) \rangle = C \operatorname{Re}\{e^{-i\Phi(t)}\} \quad | \quad \langle 2S_z \rangle = C \operatorname{Im}\{e^{-i\Phi(t)}\} \quad | \quad \langle 2S_x \rangle = 0.$$

Their derivation was straightforward and could not provide the formula for C , but they still hold, even if they only represent general equations. Thus, one gets reminded of **Eq.6.2.5**, where the ratio of the expectation values of the increasing ladder operator S_+ at times $t = T$ and $t = 0$ had, in fact, an extra complex exponential term of the Larmor angle, just like those from the equations above.

Thus, the last requirement for perfect recombination of the states ($C = 1$) is given by:

$$\Phi(T) = 2n\pi, \quad (6.4.9)$$

where n is an integer. Of course, this illustrates the precession of the spin and its periodicity—after a complete rotation, there is no difference between the original and final states.

Also interesting are the $\Phi(T) = (2n + 1)\pi$ values, as they lead to a negative expectation value of the spin. They, of course, can easily be visualized in **Fig.2.2** and represent the values on the end of the cone that lies in the x - z planes defined by $y < 0$.

6.5 A Visual Interpretation of C

In order to better interpret **Eq.6.4.7**, the behavior of C is shown in **Fig.6.1** through the terms inside the integral (**Eq.6.4.4**) that lead to the final form of C . In **Subfigure 1**, the overlap between the two wave packets is almost non-existent and shows a very poor coherence maintenance. In **Subfigures 2-4**, $\Delta x(T)$ was gradually decreased. Thus, the overlap between the two increased, as shown by the purple gaussian, which, accounting for the complex exponential form from the integral of C —which reduces it to a cosine due to its symmetry—gives the oscillatory behavior shown by the black curve. However, the full width half maximum does not change, as $\Delta x(T)$ is just a multiplicative factor for the overlap integral.

In **Subfigure 5**, the δx term was decreased, representing a squeezed state. This lead to wider gaussians. However, whether this translates to higher values of C or not is not evident. In fact, it depends on the values of the macroscopic spreads in position and momentum, as explained in **Chapter 7**.

In the last plot, the macroscopic spread in momentum was increased, leading to the emergence of more fringes on the pattern. They, however, tend to average the value of C to zero, as the integration does not occur over the modulus of the function.

Thus, one observes how the different macroscopic spreads lead to the same effective rate of decrease in coherence through two completely different phenomena.

6.6 Choosing S_+

Finally, the mathematical necessity for the choosing of the S_+ spin combination can be provided. As one moves from the Schrödinger picture to the Hermitian one, the time-evolution operators $\eta_x(t)$ and $\eta_x^\dagger(t)$ come into play. They take the form of complex exponentials of the potential energy's term, i.e., the product of the magnetic field's interaction with the spin of the states. Such a factor can thus be split into three terms, one for each Pauli spin matrix (**Eq.12.0.9**).

As the "biggest part" of the HD problem is the recombination of the states in the x direction, the $U_T(\sigma_x)$ term receives extra care and interest. Thus, the effect of a general 2×2 matrix A on the two σ_x terms from the time-evolution operators in the formula for the expectation value of the ladder operator S_+ is:

$$(-\sigma_x) \cdot A \cdot \sigma_x = \begin{pmatrix} -1 & 0 \\ 0 & 1 \end{pmatrix} \cdot \begin{pmatrix} a & b \\ c & d \end{pmatrix} \cdot \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} = \begin{pmatrix} -1 & 0 \\ 0 & 1 \end{pmatrix} \cdot \begin{pmatrix} a & -b \\ c & -d \end{pmatrix} = \begin{pmatrix} -a & b \\ c & -d \end{pmatrix}, \quad (6.6.1)$$

where its entries (a , b , c and d) are complex constants with magnitudes of 1.

In order to find a form for A , a thorough analysis of the Pauli spin matrices is in order.

Since σ_x commutes with itself and, it is only the case of the expectation value of $S_x(t)$ (**Eq.5.2.11**) when the time-evolution operators commute with themselves. Therefore, its addition into A , it would not lead to new results. Therefore, only the y and z components of the spin should be used when composing A . But since they only have components on the secondary diagonal, the only surviving entries are b and c .

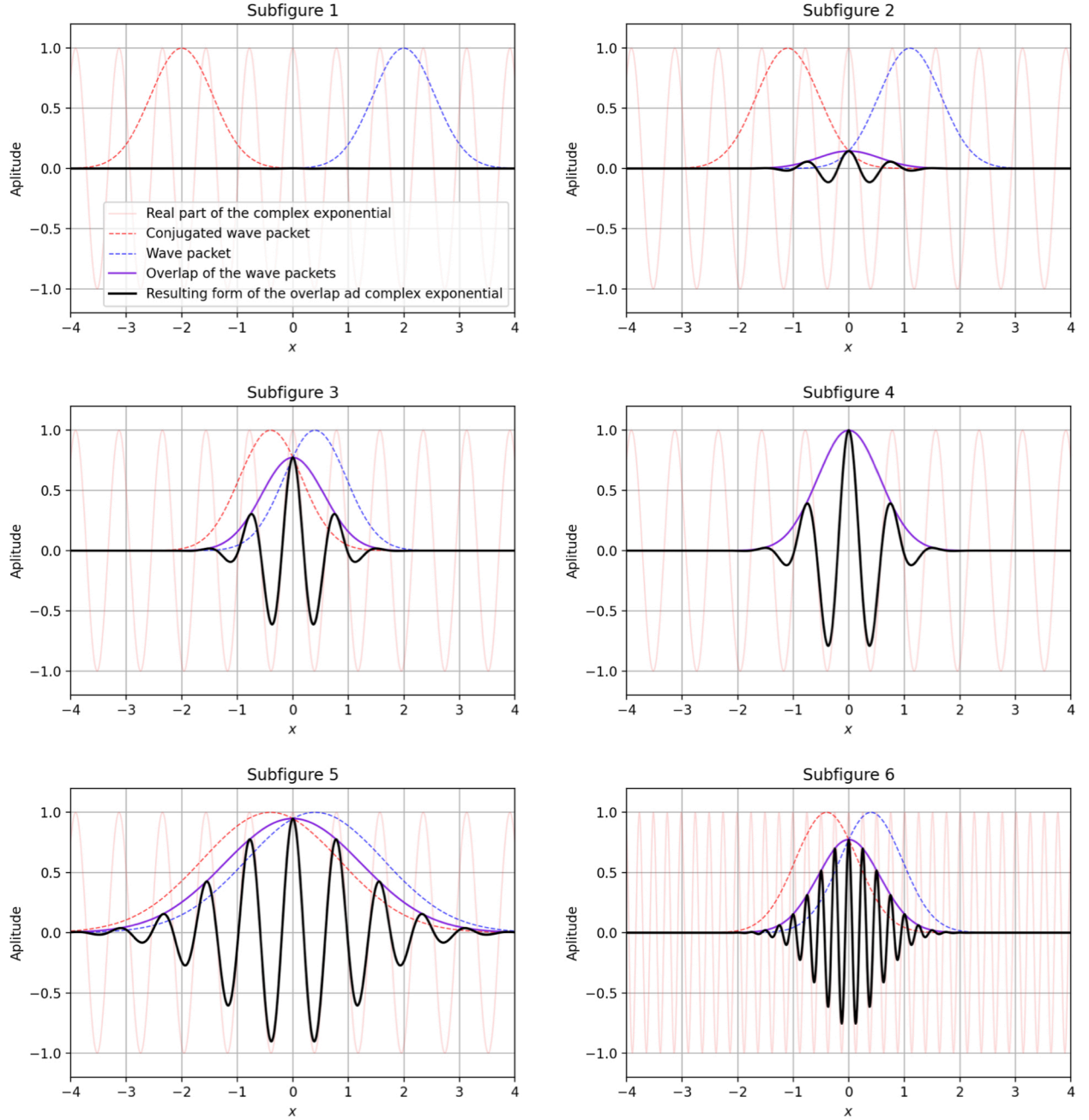


Figure 6.1: The functions constituting the integrand in **Eq.6.4.4** (or **Eq.6.4.5** with a change of variable) plotted for different variables (with red and blue dotted lines), thus showing their behavior. Their overlap is shown (thick black line) and so is the cosine behavior of the complex exponential in the integral of C (with a semi-transparent red line). In the first four subfigures, the macroscopic spread $\Delta x(T)$ was gradually decreased. In the fifth subfigure, the coefficient of x^2 in the Gaussian was decreased, while in the sixth figure, the coefficient of x was increased.

A requirement set on C is to only take one entry in the final 2×2 matrix, which, given the form of $U_t(\sigma_x)$ (**Eq.6.1.3**) and its use in **Eq.6.2.1**, implies that both σ_x matrices must interact using only one of their entries, but different ones. This translates to all but one of A 's entries being zero.

If the same entry is used, the surviving term in A —denoted by k —leads to a time-evolution of the expectation values given by:

$$U_t(+k)U_t(-k) = 1. \quad (6.6.2)$$

Thus, b must be the surviving entry.

The final trimming comes from the nature of σ_z , it must be multiplied by a strictly complex number, so when doing the final integrations to obtain C , the exponential term in the middle is not taken outside the integral directly, leaving no phase difference in the arguments of the wave function and its conjugate.

Thus, the only viable options are that of the two spin ladder operators $S_{\pm}$. Following the same procedure for determining C , but using S_- , one gets the same final result for C as in **Eq.6.4.7**, due to the integration form in **Eq.6.4.6** of Schrödinger's Gaussian. The only difference resides in the sign of the complex exponential containing the Larmor angle in **Eq.6.2.5**.

6.7 Longitudinal SG effect

One of the last sources of decoherence is the Longitudinal SG effect. When our particles have entered the magnetic field $\vec{B}(t)$, they have acquired a potential energy $-\gamma B_x(t)2S_x$, which induces a compensating change in kinetic energy $p_z \Delta v_z^{LSG}(t)$, so that:

$$\Delta v_z^{LSG}(t) = \frac{1}{p_z} \sigma_x \gamma B_x(t). \quad (6.7.1)$$

The result is an accumulated displacement of z , relative to that of a free particle. This value depends on the spin state and represents the total displacement of the wave functions in the z direction:

$$\Delta z^{LSG}(t) = \int_0^t \Delta v_z^{LSG}(t') dt' = \frac{\sigma_x}{p_z} \int_0^t \gamma B_x(t') dt' = \frac{\sigma_x}{p_z} \Phi_x(t) \quad | \quad \text{for } \left(S_x \rightarrow \pm \frac{1}{2} \right). \quad (6.7.2)$$

idkd This particular one results from the states' acquired potential energy in the magnetic field, which induces a compensating change in their kinetic energy [2]. In turn, a macroscopic spatial displacement of the states is accumulated, which must be accounted for at the end of the loop.

Chapter 7 - Squeezed States

There are cases in which one's interest is not aimed at both the position and momentum-space distributions. Such applications are high precision Michelson interferometry—used for detecting gravitational waves [23]—and, of course, the QGEM protocol. In the later, its importance is due to the gravitational interaction between the two test masses depending solely on their relative distance ($\frac{d-\Delta x}{2}$ from **Fig.2.1**). In principle, this squeezing is equivalent to quantum entanglement between the masses. In practice, detecting the squeezing is more feasible than detecting the entanglement [24].

In the context of this thesis, a squeezed state is produced when the microscopic uncertainty in position or momentum is below the $\sqrt{\frac{1}{2}}$ —value which saturates MUP in a symmetric manner—thus requiring the respective other to increase, such that no violation occurs:

$$\delta x \delta p_x = \frac{1}{2} \quad \Bigg| \quad \delta x \neq \delta p_x. \quad (7.0.1)$$

As one can see in **Fig.7.1**, this amplitude of the Gaussian decreases logarithmically with the increase of δx , while its width increases exponentially. The position expectation value $\langle x \rangle$ represents the position on the x axis of the peak. The same shape of the distribution is obtained for its momentum counterpart—by courtesy of the Fourier transform of a Gaussian being another Gaussian. Those are normalized for any values of δx and δp_x , respectively, thus naturally accommodating for the use of squeezed states.

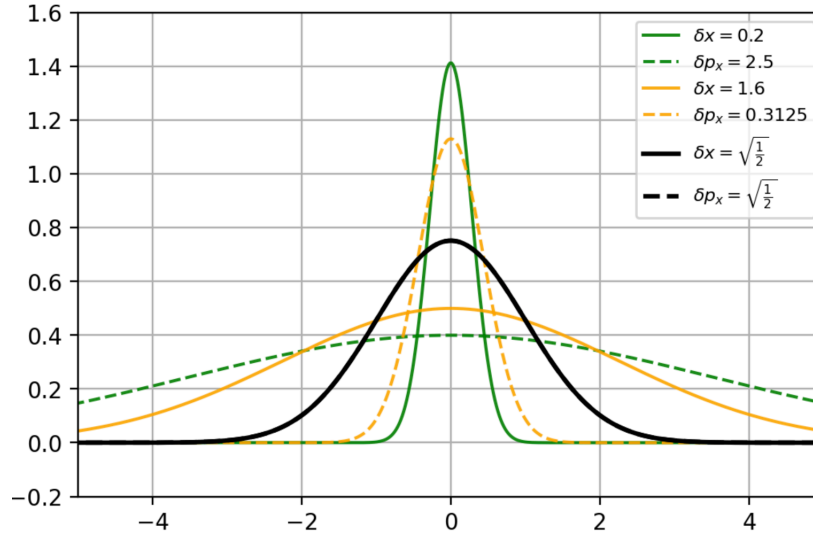


Figure 7.1: Plot of Schrödinger's Gaussian for different values of δx and $\langle x \rangle = 0$ in solid lines and the respective momentum-space distributions for $\langle p_x \rangle = 0$ in dashed lines.

Looking at **Eq.6.4.7**, one makes use of the MUP as expressed in **Eq.7.0.1** (replacing δp_x accordingly) gives:

$$C_1 = e^{-\frac{1}{2} \left[(2\delta x \Delta p_x(T))^2 + \left(\frac{\Delta x(T)}{\delta x} \right)^2 \right]}. \quad (7.0.2)$$

The derivative of C_1 with respect to δx and its (positive) root are:

$$\frac{d}{d(\delta x)} C_1 = C_1 \left(2 \frac{(\Delta x(T))^2}{(\delta x)^3} - 8 \delta x (\Delta p_x(T))^2 \right) \quad \Bigg| \quad \delta x = \sqrt{\frac{\Delta x(T)}{2 \Delta p_x(T)}}. \quad (7.0.3)$$

The behavior of this new form of the coherence measure is very interesting, as one can see in **Fig.7.2**.

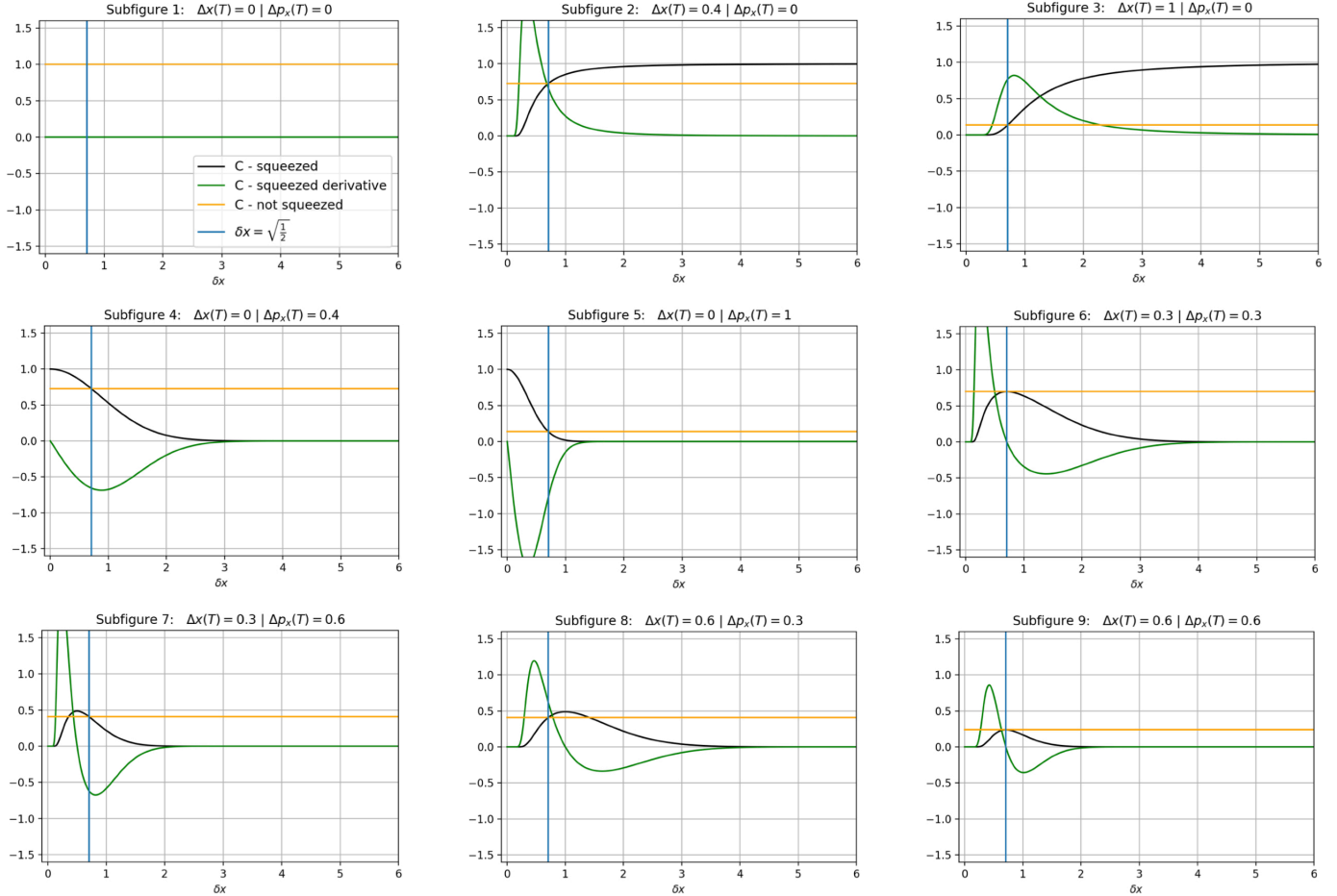


Figure 7.2: Plots of C_1 from **Eq.7.0.2** for different values of $\Delta x(T)$ and $\Delta p_x(T)$ (their exact units are not important) in terms of the δx , showing its behavior compared to its form from **Eq.6.4.7**.

In **Subfigure 1**, the case of perfect recombination is displayed. In in, C_1 and C overlap at value 1, as expected.

Subfigures 2 and **3** show the behavior in regards to changes in $\Delta x(T)$, while keeping $\Delta p_x(T) = 0$. Ass one can see, the bigger the variable, the smaller the derivative and its slope are, translating to higher values of δx required to obtain the same values of C_1 , in a linear manner. This is obvious from **Eq.7.0.2**, where the two take part in a direct division. The important observation to be made here is that, since δx now has a term to divide, C_1 starts from 0, unlike C , which, of course, is constant in each subplot and decreases exponentially with any increase in either $\Delta x(T)$ and $\Delta p_x(T)$.

However, the opposite behavior is observed in **Subfigures 4** and **5** when increasing the value of $\Delta p_x(T)$, where C_1 maintains the value of unity at its start, but exponentially decreases to 0 and does so at an increasing rate. Once, again, this is as expected from **Eq.7.0.2**, where δx depends linearly on $\Delta p_x(T)$.

Subfigures 6-9, the different combinations of position and momentum displacement are shown for the values of 0.3 and 0.6, which were chosen solely for illustrative purposes. These are the cases which one observes during an actual experiment, so they should be treated with much care and diligence. One observes in the first and last of them how the

peaks of C and C_1 intersect at the exact value of $\delta x = \sqrt{\frac{1}{2}}$, which is the one that brings symmetry into the MUP. This, of course, is as it should be, since in those cases C_1 has been effectively reduced to the form of C . However, in the other two subfigures, the combined behavior of the two observations made in the previous paragraphs is seen. It translates to a curve starting at 0 and tending towards the same value at $\delta x \rightarrow \infty$, where the original increase is much steeper for its whole duration than the symmetric points after the peak during the decrease.

The most important point to be made about the behavior of C_1 is that it leads to a very "dangerous game" where one can get much better values for the coherence measure, but also much worse ones if they do not account correctly for the magnitude of δx —taking it either too small or too large. This holds for any values of the two displacements and leaves a small "island" of the good ones between the intersections of C_1 and C , given by:

$$\delta x = \sqrt{\frac{\frac{c}{a} \pm \frac{\sqrt{c^2 - 4ab}}{a}}{2}} \quad \left| \quad a = 4b^2 = (2\Delta p_x(T))^2 \quad \left| \quad b = (\Delta x(T))^2 \quad \left| \quad c = \frac{1}{2}(a^2 + b^2) \right. \right. \right. \quad (7.0.4)$$

$$= \frac{1}{2} \left((2\Delta p_x(T))^2 + (\Delta x(T))^2 \right)^2$$

It is in between these values one must fine-tune the SGI, which requires both very high and consistent precision of the atom chips and very good knowledge of the microscopic errors in the microscopic uncertainties.

Thus, referring back to the discussion held in **Chapter 6.5**, the width of the formed wave packet whose integration (**Eq.6.4.4**) leads directly to C —and respectively to C_1 —and the number of fringes present in it now contribute in a much more complex manner.

One can conclude that (and the analogue holds), if the optimized parameter is the microscopic spread in position (δx), then it is its macroscopic counterpart ($\Delta x(T)$) which must be optimized alongside it, as it brings the peak closer to the origin and increases it exponentially while doing so. This can lead to values of C_1 much higher than those of C , as one can even more clearly see in **Fig.7.3**.

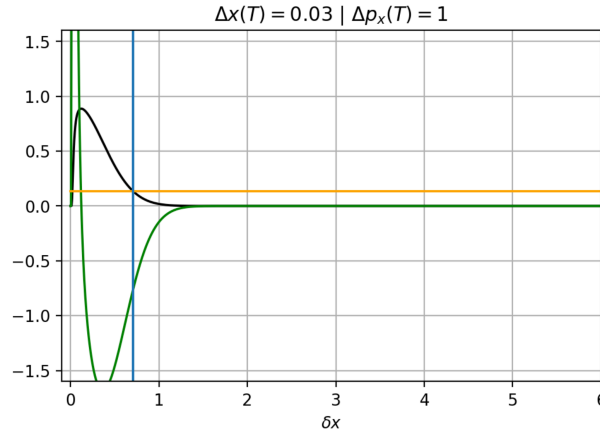


Figure 7.3: *Specific plot of C_1 from **Eq.7.0.2** for $\Delta x(T) = 0.003$ (their exact units are not important) and $\Delta p_x(T) = 1$ in terms of the δx , which were chosen in order to depict a case of good parameter fitting from the squeezed-states problem.*

Chapter 8 - More Components

8.1 The Complete Field

In this chapter, the magnetic field's components are considered in more detail by omitting the approximations made in **Chapter 4.3** and even the last simplifications in **Chapter 4.2**, while maintaining the results from **Chapters 5.4.1** and **5.4.2**, as they only depend on the spin's component which is tested ($\langle\sigma_x\rangle$).

The goal of this chapter is not primarily to solve more than what has already been done, but rather to give a different perspective to the previous work in this thesis. Thus, any additional information brought in this chapter is meant to better interpret what is already established and only give a small glimpse into what this topic's future reserves.

Thus, the used field profile is that of the second part of **Eq.4.2.10**:

$$\vec{\nabla}\vec{B}(t) = \begin{pmatrix} \frac{\partial}{\partial x}B_x(t) & 0 & \frac{1}{v}\frac{d}{dt}B_x(t) \\ 0 & -\frac{\partial}{\partial x}B_x(t) & 0 \\ \frac{1}{v}\frac{d}{dt}B_x(t) & 0 & 0 \end{pmatrix}.$$

Inserting this profile into the left (general) form of **Eq.5.2.2**, where one assumes that $\langle x \rangle = \langle p_x \rangle = 0$ and $\langle y \rangle = \langle p_y \rangle = 0$ (as discussed in **Chapter 6.4**), but leaves $\langle z \rangle$ and $\langle p_z \rangle$ open for non-zero values, the following is obtained:

$$\begin{aligned} \frac{d}{dt}\eta(t) &= i\gamma\vec{\sigma} \cdot \vec{B}(t)\eta(t) = i\gamma\vec{\sigma} \cdot \left(\vec{B}(\vec{r}(t)) + \vec{\nabla}_M\vec{B}(\vec{r}(t)) \right) \\ &= i \left\{ +\sigma_x \left[B_x(t) + \left(x + \frac{t}{m}p_x \right) \frac{\partial}{\partial x}B_x(t) + \left((z - \langle z \rangle) + \frac{t}{m}(p_z - \langle p_z \rangle) \right) \frac{1}{v}\frac{d}{dt}B_x(t) \right] \right. \\ &\quad \left. - \sigma_y \left[\left(y + \frac{t}{m}p_y \right) \frac{\partial}{\partial x}B_x(t) \right] + \sigma_z \left[\left(x + \frac{t}{m}p_x \right) \frac{1}{v}\frac{d}{dt}B_x(t) \right] \right\} \eta(t), \end{aligned} \quad (8.1.1)$$

where in the second line, the notation introduced in **Eq.4.3.3** is firstly (and lastly, thankfully) utilized. This is where the clarity of the complete mathematical description pays off.

At first glance, one might jump to the conclusion that only the $B_x(t)$ term from the first bracket induces a phase factor containing the Larmor angle. This, however, can easily be observed to be wrong, as all the partial terms which were rewritten in terms of the state's velocity in the z direction (v) are going to take part in an integration by parts with respect to time (**Eq.8.2.2**), from which such factors appear again.

By repeating the procedure in **Chapter 6.1**, one obtains:

$$\eta(T) = U_T(\sigma)\eta = [U_T(\sigma_x)U_T(\sigma_y)U_T(\sigma_z)]\eta \quad \Bigg| \quad U_T(\sigma_{x,y,z}) = e^{\int_0^T \alpha_{x,y,z}(t)dt}, \quad (8.1.2)$$

where $\alpha_{x,y,z}(t)$ represents each of the three terms in **Eq.12.0.9**.

8.2 Conserved S_x

As mentioned at the beginning of this chapter, the aim is to solidify the information already acquired and offer a different perspective. Thus, the σ_y and σ_z terms are deemed beyond the scope of this thesis and thus are ignored. The direct implication of this omission is that the x component of the spin remains conserved—**Eq.5.2.12** stays valid. The justification is that, when deducing it (the steps in **Eq.5.2.11**), the evolution operators were able to commute with σ_x , as it was also the only term in their composition.

This commutation relation does not hold anymore if other terms (σ_y or σ_z) are introduced in creation/annihilation operators, which leads to a loss of coherence. For this reason, one must reduce the components of the field in all but the x direction.

Following **Eq.8.1.2**, the form $U_T(\sigma_x)$ now adopts only differs from its previous form (second part of **Eq.6.1.3**) by the factor:

$$\begin{aligned} U'_T(\sigma_z) &= \exp \left[i\gamma\sigma_x \int_0^T \left((z - \langle z \rangle) + \frac{t}{m}(p_z - \langle p_z \rangle) \right) \frac{1}{v} \frac{d}{dt} B_x(t) dt \right] \\ &= \exp \left[i\sigma_x \left((z - \langle z \rangle) \int_0^T \gamma \frac{d}{dt} B_x(t) dt + \frac{p_z - \langle p_z \rangle}{mv} \int_0^T t \gamma \frac{d}{dt} B_x(t) dt \right) \right] \\ &= \exp \left[-i\frac{1}{2}\sigma_x \frac{p_z - \langle p_z \rangle}{\langle p_z \rangle} \Phi_x(T) \right] = \exp \left[i\frac{1}{2}\sigma_x \left(1 - \frac{p_z}{\langle p_z \rangle} \right) \Phi_x(T) \right], \end{aligned} \quad (8.2.1)$$

where one notes the extra $\langle p_z \rangle$ given by **Eq.4.2.9**. The third line made use of the physical significance of $\gamma B_x(t)$ —at both the beginning and end of the SGI, there is no magnetic field component acting on the states:

$$\int_0^T \gamma \frac{d}{dt} B_x(t) dt = \gamma B_x(t) \Big|_0^T = \gamma(B_x(T) - B_x(0)) = 0 \quad \left| \quad \int_0^T t \gamma \frac{d}{dt} B_x(t) dt = \gamma B_x(t) \Big|_0^T - \int_0^T \gamma B_x(t) dt = -\frac{1}{2} \Phi_x(T), \right. \quad (8.2.2)$$

where on the right side, in the last equality, **Eq.5.4.11** was used. The time integral (divided by v) is replaced for the spatial one in the z direction.

Of course, when performing the integration of S_+ (**Eq.6.2.1**), the argument of U'_T takes the same $\pm$ values. This gives rise to a new term in the coherence measure (see **Appendix F** for the full calculation):

$$C' = \langle \exp \left(i \frac{p_z}{\langle p_z \rangle} \Phi_x(T) \right) \rangle = \int_{-\infty}^{\infty} \Psi_0^G \left(z - \frac{\Phi_x(T)}{2\langle p_z \rangle} \right)^* \Psi_0^G \left(z + \frac{\Phi_x(T)}{2\langle p_z \rangle} \right) dz = e^{-\frac{1}{2}(\Phi_x(T) \frac{\delta p_z}{\langle p_z \rangle})^2}, \quad (8.2.3)$$

which was obtained in the same exact manner as C (using **Eq.6.4.6** with $a = \frac{1}{2(\delta p_z)^2}$ and $b = 0$), where the minimum uncertainty position wave function of a free particle in the z axis ($\Psi_0(z)$) was used. Of course, by integrating over the momentum-space wave function $\Phi_0^G(p_z)$, the same result would be obtained directly (for $a = \frac{1}{2(\delta z)^2}$ and $b = \frac{\Phi(T)}{\langle p_z \rangle}$) but then one would miss the phase addition of the argument. It is in this argument phase term where the inclusion of the $\frac{\partial}{\partial z} B_x(t)$ term shows its effect the best—the two wave-packets are spatially split by an amount proportional to the Larmor angle.

The ratio of the ladder operator's expectation values (**Eq.6.2.5**) becomes:

$$\frac{\langle S_+(T) \rangle}{\langle S_+ \rangle} = e^{-i2\Phi_x(T)} e^{-\frac{1}{2}(\Phi_x(T) \frac{\delta p_z}{\langle p_z \rangle})^2} e^{-\frac{1}{2} \left[\left(\frac{\Delta p_x(T)}{\delta p_x} \right)^2 + \left(\frac{\Delta x(T)}{\delta x} \right)^2 \right]}. \quad (8.2.4)$$

The newly introduced C' term gives rise to interesting behavior. For perfect recombination of the states ($CC' = 1$), the following condition makes its appearance:

$$\Phi_x(T) \frac{\delta p_z}{\langle p_z \rangle} = 0, \quad (8.2.5)$$

which demands a minimal value of the Larmor angle and a high one of the ratio of the uncertainty of momentum in the z direction at time $t = 0$ and the expectation value of the macroscopic momentum component in the same direction at the same time.

In the view of **Eq.6.4.9**, this leads to small values of n , thus a short integration time in **Eq.8.2.2**. In the context of the QGEM protocol, the time of integration ($T_2 + (T - T_3)$) is set to firstly and foremostly split the states in the x direction. This asks for a complex analysis and parameter fitting of the $\frac{\partial}{\partial x} B_x(t)$ and $\frac{\partial}{\partial z} B_x(t)$ components, given a field profile which contains both [14], [25], [8], thus being deemed as beyond the scope of this thesis. However, the best solution is to let the second component be zero, as it does not present any repercussions on the validity of the result, being in accordance with Maxwell's laws. This is the reason why it was reduced in **Chapters 4.2** and **4.3**. Thus, its analysis is mostly done for "educational" purposes and its revelatory component for the phenomena at hand.

Refocusing on the issue at hand, the essential point is that the Larmor angle is now part of a real exponential. It does not represent anymore just a phase term in $\langle S_+(T) \rangle$, but is part of the coherence measure itself (CC'). This behavior is not exclusive to the z component of the field. The same result can be obtained analogously for any of the field's components, given that their spatial dependence is replaced with a temporal one and their expectation value of momentum at time $t = 0$ is not zero—where the Larmor angle, of course, are dependent on the field component chosen (in this case B_x).

Thus, while the 3×1 and 1×3 entries in **Eq.8.1** share the same form, their interaction with the field is completely different, in terms of the spin's component on which they act, their spatial dependence, and, thus, also in terms of the wave-packet's component they act on.

Of course, mathematically, one can replace any integration in the making of the expectation values with the complete three-dimensional spatial integration, which returns (multiplicative) values of unity—perfect overlap—when the respective field component does not have a dependency on that axis. Nevertheless, the interpretation of the individual integrals is a whole different matter and must be treated with caution.

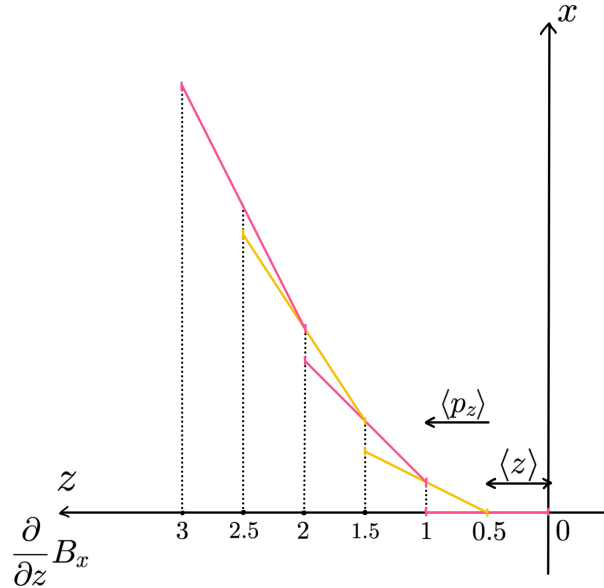


Figure 8.1: *The changes in the z wave function as the the particle moves through space due to its initial momentum. The pink and yellow lines alternate and represent the gain in position displacement on the two axis. They do not represent rotations of the wave-packets. The z axis represents both the z spatial direction and the field's gradient as labeled under it. The later was chosen specifically to fit this notation for the sake of simplicity, clarity and illustration.*

One is now remembered of the nature of the field's gradient: the magnetic field is described mathematically as a vector field. Each of its components' (spatial) partial derivatives gives rise to a different position and momentum change in the direction assigned to the component itself. Thus, while one uses the z wave function to analyze the effect above, in itself, it does not represent a spatial split in the z axis, but instead, it shows how that in the x axis is affected by the finite size of the packet, as seen for different points along the z direction. This phenomenon is depicted in **Fig.8.1**. In it, for the sake of illustration, the $\frac{\partial}{\partial x} B_x$ component has been omitted and the increase in the spatial splitting was done in a discrete manner, unlike the real integration in **Eq.8.2.3**, which, of course, is continuous.

Chapter 9 - Discussion

In this thesis, the interaction of a $\frac{1}{2}$ -spin particle inside an SGI with its magnetic field was analyzed and a formula for the coherence measure C was obtained, in accordance with that from the core material [1], [2].

The procedure through which this was achieved was detailed extensively and lead to numerous insights into the processes at hand and their implications, while being set into the context of the QGEM protocol.

The magnetic field profile used was a simplified version of those found in the majority of the papers on the different aspects of the protocol [13], despite those that deal directly with C using the one found here [9]. This goes to prove not only that more work is required in this direction, but also how the simplifications and approximations made in this thesis can still lead to results of more advanced scientific works. The $\frac{\partial}{\partial y}B_y$ component of the field is thus seemingly disregarded. According to Ampere's law, given that it is the only other non-zero gradient of the field besides $\frac{\partial}{\partial x}B_x$, it must be equal to the latter. This, of course, induces an equal splitting of the beam in the y direction—so what justifies its omission? Firstly, one can, of course, treat other aspects of the QGEM protocol in a simplified manner, just like the HD was here. A more accurate answer is that one can only be interested in the x component of the produced wave function at the end of the SGI when it comes to measurements such as C or the entanglement of states in the same direction. While this does not produce a complete description of the protocol, it is more than satisfactory in the stage it is found at. Of course, when the states are recombined, the interference pattern produced does, for the most part, propagate into the y and z directions, leaving the x one remain of interest.

A very important point to be made about the recombination of the HD is that it does not have to be happening at all. If the two wave-packets are modified—i.e., the Gaussian function is replaced inside the SGI—but they still perfectly overlap, their integral is still going to return a value of 1 for C —thus leaving it perfectly maintained. This property of the systems can be heavily exploited, most notably to abandon the symmetric structure of the SGI, which is especially useful in the case when an external potential acts more strongly on one of the states than the other—i.e., the PC and the gravitational ones [14].

As priorly discussed, the choice of the magnetic field profiles depends heavily on the requirements of the protocol—thus, its components are not free parameters. There are factors such as the avoidance of Majorana spin flips [13] which set harsh requirements on the values of the field's gradients but mostly even more so on the transition from one profile to another, which must be done adiabatically. A switching function is used in such a case to bring the step-function initially idealized field behavior to a form which can be achieved, but most importantly to make sure the experiment can be executed successfully in the first place. The existence of such factors makes the treatment of HD very hard due to the general approach it must take. But since such requirements do not usually translate to big changes in the final expressions of C , it is more advantageous to develop the theory by treating such changes as small deviations and account for them in a simplified manner. Given the number of topics and issues QGEM touches on, a "one step at a time" approach would not only be extremely inefficient but would also cause a lot of backtracking due to constantly newly discovered requirements.

Of course, the adiabatic aspect also implies extreme precision over long periods of time of the atom chips and very fine construction of components such as the BEC, the conducting plate separating the two SGIs, and most importantly, the purity of the vacuum the experiment takes place in. All of these components represent potential sources of decoherence and thus must be regarded. The imperfections caused by some of these can be interpreted as either changes in the external potential exerted on the states, while others must be regarded as stochastic noise.

As a last point to be made, coming back to the explicit HD problem, one observes that the process used to obtain the values of ± 1 in the argument $\langle [U_T(\sigma_x)]^2 \rangle$ was based on the different entries of σ_x in the time-evolution operators that interacted with that of the ladder operator S_+ . Those two components had to be different in order for the expectation

value to not reduce to unity. However, in the case where the $\frac{\partial}{\partial y}B_y$ component of the field is used, the σ_y has now to be used, whose both entries are equal. Even more worryingly, even after factoring out this time-dependence, the ladder operator inside the expectation formula is changed. To find a solution to this problem, J. Schwinger proposed in his paper [2] to replace σ_y with a combination between itself and σ_z , which can then be used to perform iterative solutions for the evolution operators. This approach is far beyond the scope of this thesis, but a complete solution is yet to be found.

Chapter 10 - Conclusion

In 1988, J. Schwinger depicted in his two papers [1], [2] the problem of spatially splitting the states of a $\frac{1}{2}$ -spin particle placed inside an SGI apparatus and attempting to recombine them. He put this process under the name of Humpty Dumpty and the measure of its "quality" as the coherence measure (C).

A particular application where his work and further improvements upon it are of utmost importance is the QGEM protocol, as described by S. Bose and A. Mazumdar in their papers [7], [9], [8], which aims to determine whether gravity is quantum in nature or not. It makes use of the LOCC and LOQC methods, thus proposing the observation of the (possible) gravitational entanglement of two spatially split $\frac{1}{2}$ -spin states in two closely placed SGIs.

The general experimental setup was described, leading to an argument for the importance of obtaining coherent states at the end of the two SGIs. This was used to give an early definition of the coherence measure and its experimental implications, while setting x as the axis of splitting. The HD problem was then restricted to a single SGI, thus excluding any external interactions for a simplified treatment.

Following the simple argument of a $\frac{1}{2}$ -spin particle in a homogenous magnetic field pointing in the x direction, a clearer image of the coherence measure and how it acts on the spin's components was depicted. At the same time, the use of the increasing spin ladder operator as one of the primary tools in the construction of theory was justified, as the real and complex parts of its expectation value represent those of the spin's y and, respectively, z components. Thus, the coherence measure was defined as the ratio between the expectation values of the ladder operator at the beginning and end of the SGI.

The complete transition to the quantum aspect of the HD was made through the introduction of the system's Hamiltonian and the time-evolution operators used to replace the Schrödinger picture with the Hermitian one. This allowed for a cleaner analysis and deduction of the equations at hand.

In order to maximize the coherence measure, restrictions were placed on the field profile. An extensive analysis of how each such curtailment must be set and its implications followed by heavily using Maxwell's equations. This led to the survival of only two field components— $B_x(t)$ and $\frac{\partial}{\partial x}B_x(t)$, and the omission of $\frac{\partial}{\partial y}B_y(t)$ for the scope of this thesis. The latter requires further analysis by different methods, such as an iterative method.

The interaction of the spin with the Hamiltonian led to a linear time dependence on the momentum operator at time $t = 0$ of the field's argument, which permitted its replacement with the time coordinate. The mechanism behind this was described as the lack of interaction between the spin's null states and the fields' components depending on it.

A clear and direct path of finding the explicit expression of C followed, where the distinction between the Gaussian minimum uncertainty wave-packets and the stationary wave functions which form them was established. This permitted the addition of the accumulated momentum and position separations in the arguments of the two Gaussians. The produced packets displayed the result of all of the processes inside the SGI mathematically through the partition of these displacements between the two packets.

From here, the final integration of the modified wave-packets followed, leading directly to the same formula of C . Its expression revealed its explicit—inverse—exponential behavior on each of the separations—with a minus term—which were divided by their respective microscopic errors. While this result in itself is identical to that from J. Schwinger's work and thus could have been directly picked up from there, the visualization and analysis of the individual steps and their implications paid off, as seen by the many new points brought up throughout this thesis, culminating in the **Discussion** section, where the HD problem was able to be easily reanalyzed in the context of the QGEM.

After this original formula of C was obtained, the mathematical need for the use of S_+ was expressed, followed by a short description of the LSG effect.

Further analysis of a slightly more complicated magnetic field followed, where the $\frac{\partial}{\partial z}B_x$ gradient component was used

to provide a different perspective to the already obtained results and bring to the surface the z component of the wave-packet, which was also assumed to take the form of Schrödinger's Gaussian. This resulted in a new term in the formula of the coherence measure (C'), most notably containing Larmor angle's x component. While seemingly deceptive, the result accounted, in fact, for the displacement in the x direction due to the finite size of the packet, as seen for different points along the y direction.

This approach to the HD problem of a $\frac{1}{2}$ -spin particle in an SGI provided a deep insight into the mechanism behind the results in [1] and [2], while adapting to the QGEM protocol, for which it is of utmost importance. While it can lead to concrete results in the protocol even in this early form, more work by theorists needs to be put into the HD and all the other complications present in the protocol, and so by experimentalists into developing better, more precise tools. Nevertheless, it shall all be worth the effort when the day when the final verdict on the graviton will be given.

Chapter 11 - Acknowledgements

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Secondly, I wish to thank the University of Groningen for the tools, education, and facilities they have provided, which allowed me and my fellow colleagues to finish our Bachelor's degrees, especially given the difficult conditions imposed by the ongoing pandemic.

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Appendices

Appendix A

The argumentation of **Eq.2.1.4** is surprisingly tricky. Following the argument in [26], for a wave-packet moving with (group) velocity v_c to first approximation, the Taylor expansion dispersion relation can be linear—non-dispersive—as well:

$$\omega(p_x) = \omega(\langle p_x \rangle) + (p_x - \langle p_x \rangle)\omega'(\langle p_x \rangle) + \frac{1}{2}(p_x - \langle p_x \rangle)^2\omega''(\langle p_x \rangle) = \langle p_x \rangle v_p + (p_x - \langle p_x \rangle)v_g + \frac{1}{2}(p_x - \langle p_x \rangle)^2\Gamma \quad (12.0.1)$$

where v_p is the phase velocity and Γ is the broadening rate.

Looking back at **Eq.2.3.1**:

$$\Psi_0^G(x) = (2\pi)^{-\frac{1}{4}}(\delta x)^{-\frac{1}{2}}e^{i\langle p_x \rangle(x - \langle x \rangle)}e^{-\frac{1}{4}\left(\frac{x - \langle x \rangle}{\delta x}\right)^2}, \quad (12.0.2)$$

one can write the Fourier transform as:

$$\tilde{\Psi}_0^G(p_x(t)) = (2\pi)^{-\frac{3}{4}}(\delta x)^{\frac{1}{2}}e^{-[\delta x(p_x - \langle p_x \rangle)]^2}e^{i\langle x \rangle(p_x - \langle p_x \rangle)}, \quad (12.0.3)$$

from which the time-dependent wave-packet is obtained:

$$\Psi_0(x(t), t) = (2\pi)^{-\frac{3}{4}}(\delta x)^{\frac{1}{2}} \int_{-\infty}^{\infty} e^{ip_x x - \omega(p_x)t} e^{-[\delta x(p_x - \langle p_x \rangle)]^2} e^{i\langle x \rangle(p - \langle p_x \rangle)} dp_x \quad (12.0.4)$$

In the above equation, the separate t in the argument represents an actual time dependence of the packet, which represents the packet's evolution in time and not just its positioning. Keeping **Eq.12.0.1** in mind, the exponential is still quadratic in p_x —still a Gaussian—and thus the inverse Fourier transform can be performed, resulting in a new Gaussian function:

$$\begin{aligned} \Psi_0(x(t), t) &= (2\pi)^{-\frac{1}{4}}(\delta x)^{-\frac{1}{2}}e^{\frac{1}{4}\left(\frac{x - (\langle x \rangle + v_g t)}{\sqrt{(\delta x)^2 - i\frac{1}{2}\Gamma t}}\right)^2} e^{i\langle p_x \rangle x} e^{-i\langle p_x \rangle t(v_g - v_p)} \\ &= (2\pi)^{-\frac{1}{4}}(\delta x)^{-\frac{1}{2}}e^{\frac{1}{4}\left(\frac{x - (\langle x \rangle + v_g t)}{\delta x \sqrt{1 - i\frac{1}{2(\delta x)^2}\Gamma t}}\right)^2} e^{i\langle p_x \rangle x} e^{-i\langle p_x \rangle t(v_g - v_p)} \\ &= (2\pi)^{-\frac{1}{4}}(\delta x)^{-\frac{1}{2}}e^{\frac{1}{4}\left(\frac{x - (\langle x \rangle + v_g t)}{(\delta x(t))^2}\right)^2} e^{i\phi(x, t)}, \end{aligned} \quad (12.0.5)$$

where the following substitutions were used:

$$\delta x(t) = \delta x \sqrt{1 + \left(\frac{\Gamma t}{2(\delta x)^2}\right)^2} \quad \left| \quad \phi(x, t) = \langle p_x \rangle x - \langle p_x \rangle t(v_g - v_p) - \frac{t\Gamma}{\left(\frac{t\Gamma}{2}\right)^2 + (\delta x)^4}. \right. \quad (12.0.6)$$

From Schrödinger's Gaussian (**Eq.2.3.1**) one deduces that:

$$\omega(p_x) = \frac{p_x^2}{2m} \quad \Bigg| \quad \Gamma = \frac{1}{m}, \quad (12.0.7)$$

which, when plotted into the first part of **Eq.12.0.6**, give the final formula for the time dependence of the (microscopic) position spread:

$$\delta x(t) = \delta x \sqrt{1 + \left(\frac{\Gamma t}{2(\delta x)^2} \right)^2} = \delta x \sqrt{1 + \left(\frac{t}{m} \right)^2 \left(\frac{1}{2(\delta x)^2} \right)^2} = \delta x \sqrt{1 + \left(\frac{t}{m} \right)^2 \left(\frac{\delta p_x}{\delta x} \right)^2} = \sqrt{(\delta x)^2 + \left(\frac{t}{m} \right)^2 (\delta p_x)^2}, \quad (12.0.8)$$

which is exactly the (square root of the) **Eq.2.1.4**.

Appendix B

$$\begin{aligned} \langle \eta^\dagger(t) \frac{\vec{p}(t)^2}{2m} \eta(t) \rangle &= \frac{1}{2m} \langle \eta^\dagger(t) \left[\vec{p} + \int_0^t \gamma \vec{\nabla} \vec{B}(\vec{r}(t')) \circ (\eta^\dagger(t') \vec{\sigma} \eta(t')) dt' \right]^2 \eta(t) \rangle \\ &= \frac{1}{2m} \left[\langle \eta^\dagger(t) (\langle \vec{p} \rangle^2 \eta(t)) \rangle + \langle \eta^\dagger(t) \left(2\vec{p} \int_0^t \gamma \vec{\nabla} \vec{B}(\vec{r}(t')) \circ (\eta^\dagger(t') \vec{\sigma} \eta(t')) dt' \right) \eta(t) \rangle \right. \\ &\quad \left. + \langle \eta^\dagger(t) \left(\int_0^t \gamma \vec{\nabla} \vec{B}(\vec{r}(t')) \circ (\eta^\dagger(t') \vec{\sigma} \eta(t')) dt' \right) \eta(t) \rangle \right] \\ &= \frac{1}{2m} \langle \eta^\dagger(t) \vec{p}^2 \eta(t) \rangle = \frac{1}{2m} \vec{p}^2 \langle \eta^\dagger(t) \eta(t) \rangle = \frac{1}{2m} \vec{p}^2 \end{aligned} \quad (12.0.9)$$

In going from the second to the third line, one makes use of the null spin state $\eta(t)$ which acts on the spin $\vec{\sigma}$ (see **Chapter 5.4.1**). Of course, the *bra ket* notation can be dropped in **Eq.5.2.1**.

Appendix C

$$\begin{aligned} \eta_x(t) &= \eta_x \cdot e^{i\gamma\sigma_x \int_0^t [B_x(t') + (x + \frac{t'}{m} p_x) \frac{\partial}{\partial x} B_x(t')] dt'} = \eta_x \cdot e^{i\gamma\sigma_x [\int_0^t B_x(t') dt' + \int_0^t (x + \frac{t'}{m} p_x) \frac{\partial}{\partial x} B_x(t') dt']} \\ &= \eta_x \left[e^{i\gamma\sigma_x \int_0^t B_x(t') dt'} e^{i\gamma\sigma_x \int_0^t (x + \frac{t'}{m} p_x) \frac{\partial}{\partial x} B_x(t') dt'} \right] = \eta_x \left[e^{i\sigma_x \frac{1}{2} \Phi_x(t)} e^{i\gamma\sigma_x (x \int_0^t \frac{\partial}{\partial x} B_x(t') dt' - p_x \int_0^t -(\frac{t'}{m}) \frac{\partial}{\partial x} B_x(t') dt')} \right] \\ &= \eta_x \left[e^{i\sigma_x \frac{1}{2} \Phi_x(t)} e^{i\sigma_x (x \Delta p_x(t) - p_x \bar{\Delta} x(t))} \right] = \eta_x U_t(\sigma_x) \end{aligned} \quad (12.0.10)$$

Appendix D

$$\begin{aligned}
C &= \int_{-\infty}^{\infty} \Psi_0(x)^* e^{-2i(x\Delta p_x(T) - p_x\bar{\Delta}x(T))} \Psi_0(x) dx \\
&= \int_{-\infty}^{\infty} \left(A e^{i(xp_x - Et)} \right)^* e^{-2i(x\Delta p_x(T) - p_x\bar{\Delta}x(T))} \left(A e^{i(xp_x - Et)} \right) dx \\
&= \int_{-\infty}^{\infty} \left(A e^{-ip_x\bar{\Delta}x(T)} e^{i(xp_x - Et)} \right)^* e^{-2ix\Delta p_x(T)} \left(A e^{ip_x\bar{\Delta}x(T)} e^{i(xp_x - Et)} \right) dx \\
&= \int_{-\infty}^{\infty} \left(A e^{i[p_x(x - \bar{\Delta}x(T)) - Et]} \right)^* e^{-2ix\Delta p_x(T)} \left(A e^{i[p_x(x + \bar{\Delta}x(T)) - Et]} \right) dx \\
&= \int_{-\infty}^{\infty} \Psi_0(x - \bar{\Delta}x(T))^* e^{-2ix\Delta p_x(T)} \Psi_0(x + \bar{\Delta}x(T)) dx
\end{aligned} \tag{12.0.11}$$

$$\begin{aligned}
C &= \int_{-\infty}^{\infty} \Psi_0(p_x)^* e^{-2i(x\Delta p_x(T) - p_x\bar{\Delta}x(T))} \Psi_0(p_x) dp_x \\
&= \int_{-\infty}^{\infty} \left(A e^{-i(xp_x + Et)} \right)^* e^{-2i(x\Delta p_x(T) - p_x\bar{\Delta}x(T))} \left(A e^{-i(xp_x + Et)} \right) dp_x \\
&= \int_{-\infty}^{\infty} \left(A e^{ix\Delta p_x(T)} e^{-i(xp_x + Et)} \right)^* e^{2ip_x\bar{\Delta}x(T)} \left(A e^{-ix\Delta p_x(T)} e^{-i(xp_x + Et)} \right) dp_x \\
&= \int_{-\infty}^{\infty} \left(A e^{-i[x(p_x - \Delta p_x(T)) + Et]} \right)^* e^{-2ix\Delta p_x(T)} \left(A e^{-i[x(p_x + \Delta p_x(T)) + Et]} \right) dp_x \\
&= \int_{-\infty}^{\infty} \Phi_0(p_x - \Delta p_x(T))^* e^{-2ix\Delta p_x(T)} \Phi_0(p_x + \Delta p_x(T)) dp_x
\end{aligned} \tag{12.0.12}$$

Appendix E

$$\begin{aligned}
C &= \int_{-\infty}^{\infty} \Psi_T(x - \Delta x(T))^* e^{-2ix\Delta p_x(T)} \Psi_T(x + \Delta x(T)) dx \\
&= (2\pi)^{-\frac{1}{2}} (\delta x)^{-1} \int_{-\infty}^{\infty} e^{-\frac{1}{4} \left(\frac{x - \Delta x(T)}{\delta x} \right)^2} e^{-2ix\Delta p_x(T)} e^{-\frac{1}{4} \left(\frac{x + \Delta x(T)}{\delta x} \right)^2} dx \\
&= (2\pi)^{-\frac{1}{2}} (\delta x)^{-1} \int_{-\infty}^{\infty} e^{-\frac{1}{2} \frac{x^2 + (\Delta x(T))^2}{(\delta x)^2}} e^{-2ix\Delta p_x(T)} dx \\
&= (2\pi)^{-\frac{1}{2}} (\delta x)^{-1} e^{-\frac{1}{2} \left(\frac{\Delta x(T)}{\delta x} \right)^2} \int_{-\infty}^{\infty} e^{-\frac{1}{2} \left(\frac{x}{\delta x} \right)^2} e^{-2ix\Delta p_x(T)} dx \\
&= (2\pi)^{-\frac{1}{2}} (\delta x)^{-1} e^{-\frac{1}{2} \left(\frac{\Delta x(T)}{\delta x} \right)^2} \int_{-\infty}^{\infty} e^{-\frac{1}{2(\delta x)^2} x^2} e^{i(-2\Delta p_x(T))x} dx
\end{aligned} \tag{12.0.13}$$

$$\begin{aligned}
C &= \int_{-\infty}^{\infty} \Psi_T(p_x - \Delta p_x(T))^* e^{2ip_x \Delta x(T)} \Psi_T(p_x + \Delta p_x(T)) dp_x \\
&= (2\pi)^{-\frac{1}{2}} (\delta p_x)^{-1} \int_{-\infty}^{\infty} e^{-\frac{1}{4} \left(\frac{p_x - \Delta p_x(T)}{\delta p_x} \right)^2} e^{2ip_x \Delta x(T)} e^{-\frac{1}{4} \left(\frac{p_x + \Delta p_x(T)}{\delta p_x} \right)^2} dp_x \\
&= (2\pi)^{-\frac{1}{2}} (\delta p_x)^{-1} \int_{-\infty}^{\infty} e^{-\frac{1}{2} \frac{p_x^2 + (\Delta p_x(T))^2}{(\delta p_x)^2}} e^{2ip_x \Delta x(T)} dp_x \\
&= (2\pi)^{-\frac{1}{2}} (\delta p_x)^{-1} e^{-\frac{1}{2} \left(\frac{\Delta p_x(T)}{\delta p_x} \right)^2} \int_{-\infty}^{\infty} e^{-\frac{1}{2} \left(\frac{p_x}{\delta p_x} \right)^2} e^{2ip_x \Delta x(T)} dp_x \\
&= (2\pi)^{-\frac{1}{2}} (\delta p_x)^{-1} e^{-\frac{1}{2} \left(\frac{\Delta p_x(T)}{\delta p_x} \right)^2} \int_{-\infty}^{\infty} e^{-\frac{1}{2(\delta p_x)^2} p_x^2} e^{i(2\Delta x(T))p_x} dp_x
\end{aligned} \tag{12.0.14}$$

Appendix F

$$\begin{aligned}
\left\langle \exp \left[i\Phi(T) \frac{p_y - \langle p_y \rangle}{\langle p_y \rangle} \right] \right\rangle &= \left\langle \exp \left[i\Phi(T) \left(\frac{p_y}{\langle p_y \rangle} - \frac{\langle p_y \rangle}{\langle p_y \rangle} \right) \right] \right\rangle = \left\langle \exp \left[i\Phi(T) \left(\frac{p_y}{\langle p_y \rangle} - 1 \right) \right] \right\rangle \\
&= \left\langle \exp \left(i\Phi(T) \frac{p_y}{\langle p_y \rangle} \right) \cdot \exp(-i\Phi(T)) \right\rangle = \exp(-i\Phi(T)) \left\langle \exp \left(i\Phi(T) \frac{p_y}{\langle p_y \rangle} \right) \right\rangle \\
&= e^{-i\Phi(T)} (2\pi)^{-\frac{1}{2}} (\delta p_y)^{-1} \int_{-\infty}^{\infty} e^{-\frac{1}{4} \left(\frac{p_y}{\delta p_y} \right)^2} e^{i\Phi(T) \frac{p_y}{\langle p_y \rangle}} e^{-\frac{1}{4} \left(\frac{p_y}{\delta p_y} \right)^2} dp_y \\
&= e^{-i\Phi(T)} (2\pi)^{-\frac{1}{2}} (\delta p_y)^{-1} \int_{-\infty}^{\infty} e^{-\frac{1}{2} \left(\frac{p_y}{\delta p_y} \right)^2} e^{i\Phi(T) \frac{p_y}{\langle p_y \rangle}} dp_y \\
&= e^{-i\Phi(T)} (2\pi)^{-\frac{1}{2}} (\delta p_y)^{-1} \int_{-\infty}^{\infty} e^{-(2(\delta p_y)^2)^{-1} p_y^2} e^{i \frac{\Phi(T)}{\langle p_y \rangle} p_y} dp_y \\
&= e^{-i\Phi(T)} (2\pi)^{-\frac{1}{2}} (\delta p_y)^{-1} e^{-\frac{\left(\frac{\Phi(T)}{\langle p_y \rangle} \right)^2}{4(2(\delta p_y)^2)^{-1}}} \sqrt{\pi} \left| \sqrt{\frac{\pi}{(2(\delta p_y)^2)^{-1}}} \right| \\
&= e^{-i\Phi(T)} e^{-\frac{\left(\frac{\Phi(T)}{\langle p_y \rangle} \right)^2}{4(2(\delta p_y)^2)^{-1}}} = e^{-i\Phi(T)} e^{-\frac{1}{2} \left(\Phi(T) \frac{p_y}{\langle p_y \rangle} \right)^2}
\end{aligned} \tag{12.0.15}$$

The above equation was obtained in the same exact manner as its **Eq.6.4.7**—that is, using **Eq.6.4.6** with $a = \frac{1}{2(\delta p_z)^2}$ and $b = 0$, but using the minimum uncertainty position wave function of a free particle in the z axis ($\Psi_0(z)$).

Appendix G

In **Chapter 6.3**, the process by which the phases produced inside the SGI are assimilated inside the Gaussian wave function's argument was explained. The reasoning was, on short, that they must first act on the wave function of free particles, which follow the momentum/position distribution that forms the Gaussian. It is at this point when the ordering of the above was set: one must start with the wave functions of free particles and only then can those be integrated (and the momentum and position distributions, respectively, be introduced) to form the wave-packets.

The following is a calculation of the wrong procedure when one starts with the Gaussian wave-packets. However, it does not stop in the point where "it should" (when the phases could not be introduced into the packet's integration, step denoted by $\neq$), but bypasses it, showing exactly what this procedure's downfall is:

$$\begin{aligned}
C &= \int_{-\infty}^{\infty} \Psi_0^\dagger(x) e^{2ip_x \bar{\Delta}x(T)} e^{-2ix\Delta p_x(T)} \Psi_0(x) dx = \int_{-\infty}^{\infty} \left[\int_{-\infty}^{\infty} \frac{1}{\sqrt{2\pi}} \left(\frac{2(\delta x)^2}{\pi} \right)^{\frac{1}{4}} e^{-(\delta x)^2(p_x - \langle p_x \rangle)^2} e^{ip_x x} dp_x \right]^\dagger \\
&\quad e^{2ip_x \bar{\Delta}x(T)} e^{-2ix\Delta p_x(T)} \left[\int_{-\infty}^{\infty} \frac{1}{\sqrt{2\pi}} \left(\frac{2(\delta x)^2}{\pi} \right)^{\frac{1}{4}} e^{-(\delta x)^2(p_x - \langle p_x \rangle)^2} e^{ip_x x} dp_x \right] dx \\
&= \frac{\delta x}{\sqrt{2\pi^3}} \int_{-\infty}^{\infty} \left[\int_{-\infty}^{\infty} e^{-(\delta x)^2(p_x - \langle p_x \rangle)^2} e^{ip_x x} dp_x \right]^\dagger e^{2ip_x \bar{\Delta}x(T)} e^{-2ix\Delta p_x(T)} \left[\int_{-\infty}^{\infty} e^{-(\delta x)^2(p_x - \langle p_x \rangle)^2} e^{ip_x x} dp_x \right] dx \\
&= \frac{\delta x}{\sqrt{2\pi^3}} \int_{-\infty}^{\infty} \left[e^{-ip_x \bar{\Delta}x(T)} \int_{-\infty}^{\infty} e^{-(\delta x)^2(p_x - \langle p_x \rangle)^2} e^{ip_x x} dp_x \right]^\dagger e^{-2ix\Delta p_x(T)} \left[e^{ip_x \bar{\Delta}x(T)} \int_{-\infty}^{\infty} e^{-(\delta x)^2(p_x - \langle p_x \rangle)^2} e^{ip_x x} dp_x \right] dx \\
&\neq \frac{\delta x}{\sqrt{2\pi^3}} \int_{-\infty}^{\infty} \left[\int_{-\infty}^{\infty} e^{-ip_x \bar{\Delta}x(T)} e^{-(\delta x)^2(p_x - \langle p_x \rangle)^2} e^{ip_x x} dp_x \right]^\dagger e^{-2ix\Delta p_x(T)} \left[\int_{-\infty}^{\infty} e^{ip_x \bar{\Delta}x(T)} e^{-(\delta x)^2(p_x - \langle p_x \rangle)^2} e^{ip_x x} dp_x \right] dx \\
&= \frac{\delta x}{\sqrt{2\pi^3}} \int_{-\infty}^{\infty} \left[\int_{-\infty}^{\infty} e^{-(\delta x)^2(p_x - \langle p_x \rangle)^2} e^{ip_x(x - \bar{\Delta}x(T))} dp_x \right]^\dagger e^{-2ix\Delta p_x(T)} \left[\int_{-\infty}^{\infty} e^{-(\delta x)^2(p_x - \langle p_x \rangle)^2} e^{ip_x(x + \bar{\Delta}x(T))} dp_x \right] dx \\
&= \int_{-\infty}^{\infty} \Psi_0^\dagger(x - \bar{\Delta}x) e^{-2ix\Delta p_x(T)} \Psi_0(x + \bar{\Delta}x) dx.
\end{aligned} \tag{12.0.16}$$

This procedure's incorrectness now becomes clear and its exact failing point confirms argument made in **Chapter 6.3**.